

Torddis: A Real-Time IoT and AI-Based System to Support At-Home Learning and Mitigate Parental Absenteeism

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Abstract

Parental absenteeism during out-of-school study hours has intensified in recent years, undermining the academic performance and self-regulation of primary school children. This paper introduces Torddis, a real-time system that combines Internet of Things (IoT) hardware and Artificial Intelligence (AI) models to support at-home learning when tutors cannot provide continuous supervision. Torddis integrates an ESP32-CAM/NodeMCU device with a mobile application; it detects facial expressions indicative of inattention, unauthorised objects on the desk, and signs of drowsiness, and returns visual and auditory alerts that promote self-correction. The system was designed and validated following the Test-Driven Development Methodology for IoT Systems (TDDM4IoTS). A pilot deployment was carried out in twelve families over one week; tutors evaluated the prototype through the System Usability Scale (SUS) and a purpose-built seven-item Likert questionnaire on perceived effectiveness. Torddis obtained a mean SUS score of 81.46% (SD = 11.65), corresponding to a “Good” usability level. Statistical analyses (Shapiro–Wilk, ANOVA, Kruskal–Wallis, Pearson and Spearman correlations) revealed a strong positive association between the use of the device and tutor-perceived improvements in children’s concentration and self-regulation. The findings support the pedagogical value of AI-enabled IoT tools for autonomous learning at home and outline avenues for larger and longer-term evaluations.

Keywords: Educational technology, Internet of Things, Artificial intelligence, Self-regulated learning, Home-based learning, Usability evaluation

1 Introduction

Over the last decade, accelerated lifestyles and increasing workloads have contributed to a rise in parental absenteeism during the hours designated for schoolwork at home, particularly among primary school students. This situation has underscored the need for technological solutions to compensate for the lack of supervision (Ketsman, Droog, & Qazi, 2025; Seidu, Arthur-Holmes, Agbaglo, & Ahinkorah, 2022).

The absence of parental support directly impacts the academic performance of young learners. Teachers have observed a decline in indicators related to performance and discipline, which may adversely affect students' self-esteem and other psychosocial dimensions. Furthermore, the literature emphasizes that emotional support in the home is a fundamental pillar for maintaining children's focus on their studies (Seidu et al., 2022).

Emotional interference not only influences educational outcomes but also significantly affects other aspects of students' lives (Åke, Hartelius, Jakola, & Antonsson, 2023). In this context, a family environment that fosters positive communication and genuine interaction can enhance students' well-being and social skills (Navarro, Larrañaga, Yubero, & Vllora, 2024). However, to effectively address the challenges associated with attention and emotional regulation, it is crucial to incorporate technological tools that integrate Artificial Intelligence (AI) and the Internet of Things (IoT). These technologies enable the monitoring and measurement of children's distractions, providing essential data to support decision-making processes aimed at improving attention and learning outcomes (Alvear-Puertas, Rosero-Montalvo, Peluffo-Ordóñez, & Pijal-Rojas, 2017; Berrezueta-Guzman, Pau, Martín-Ruiz, & Máximo-Bocanegra, 2021).

The use of AI- and IoT-based technologies not only contributes to the improvement of academic performance but also promotes self-regulation skills in students, which are essential for their cognitive and socio-emotional development (Mohamed & Lamia, 2018). The feedback provided by such technologies helps students identify their distractions and take corrective actions, strengthening their ability to manage time and concentrate on academic tasks. Consequently, technological tools emerge as valuable resources to complement the positive influence of the family environment and to foster self-regulated learning, creating a comprehensive approach that encompasses both educational and personal development (Thinyane & Masikisiki, 2016).

Students' ability to concentrate is a crucial factor in their capacity to learn and apply knowledge delivered by educators. In a study conducted by Terraza Arciniegas et al. (2022), a system was proposed to monitor students' attention in an online learning environment using computer vision algorithms. The authors implemented a Recurrent Neural Network (RNN) to identify facial landmarks, detecting whether a student maintained a frontal position toward the device used to attend virtual classes. Their study reveals some limitations present in current proposals.

On the other hand, the work of Berrezueta-Guzman et al. (2021) introduced a robotic system designed to interact with children diagnosed with Attention Deficit Hyperactivity Disorder (ADHD), helping them correct unhealthy habits and inappropriate behaviors associated with this condition. Although effective within its specific scope, this system is exclusively oriented toward children with ADHD. Its generalized

application may lead to disruptions for children without this condition, and it lacks notification, alert, or alarm functionalities to inform tutors about potential incidents.

To promote active tutor participation and enhance children’s emotional well-being, this paper introduces Torddis¹, a technological solution designed to monitor student behavior while they autonomously perform academic tasks at home, notifying tutors of any significant event and conveying to students the feeling that they are not alone. In addition, Torddis is capable of playing sounds, including personalized messages according to user preferences.

To achieve these objectives, Torddis employs advanced computer vision algorithms and deep learning techniques. The system includes a mobile application for tutors and an IoT device that analyzes children’s behavior while they complete school-work independently. Developed following the TDDM4IoTS methodology [reference anonymised], Torddis features a holistic approach, enabling tutors to monitor and respond in real time to signs of inattention. Key functionalities include the detection of facial expressions associated with core emotions, identification of drowsiness, and recognition of inappropriate use of distractive objects during academic activities. These features allow tutors to encourage their children to remain focused through the use of visual and auditory reinforcement cues (Al-Gburi, Al-Sammak, Marghescu, & Oprea, 2023; Enadula, Enadula, & Burri, 2021; Terraza Arciniegas et al., 2022).

Torddis is designed to strengthen metacognitive skills such as time management and sustained attention, which are essential components of modern autonomous learning. Its emphasis on proactive supervision addresses the need to support children during a critical stage of development, providing tutors with a valuable tool that complements traditional supervision at home and contributes to improving educational outcomes within this population.

This article is organized as follows: Section 2 examines the pedagogical dimension of the Torddis system. Section 3 presents a review of the most relevant related works published in prestigious international journals over the last ten years. Section 4 provides a detailed description of the materials and methods used during the system’s development, including the evaluation processes. Section 5 outlines the methodological design of the study and the experimental validation process. Section 6 discusses the main findings and their implications, while Section 7 presents a detailed statistical analysis of the study. Finally, Section 8 concludes the article and suggests potential future research directions.

2 Pedagogical Dimension of Torddis

Torddis not only provides advanced technological solutions for monitoring students’ concentration but also plays a fundamental role in promoting key pedagogical skills such as self-regulation, self-reflection, and autonomous learning. By offering immediate and personalized feedback, the system supports the development of effective strategies that enhance time management and attention—competencies that are essential for modern learning (Loh, Kraaykamp, & van Hek, 2025).

¹From Latin *Torqueo discerno discipulus*, meaning “detection of deviation or change in the student’s attention.”

2.1 Fostering Self-Regulation and Self-Reflection

Self-regulation, defined as students' ability to manage and assess their own learning process, is an essential competence for educational development. Torddis actively fosters this skill through an immediate feedback system that detects real-time distractions, such as changes in facial expressions, unauthorized objects (distractors), and signs of drowsiness. The system's visual and auditory alerts help students redirect their attention to academic tasks, thereby improving their focus and reinforcing self-reflection to identify and overcome learning barriers (Ackermans, Bakker, van Loon, Kral, & Camp, 2025; Li, Lowell, Wang, & Li, 2024).

This active detection capability enables students to develop key metacognitive skills by recognizing and correcting their own distractive behaviors. The system design aligns with theoretical models such as Zimmerman's model of self-regulation (Zimmerman & Schunk, 2013) and Tough's model of autonomous learning (Tough, 1985), both of which emphasize the importance of immediate feedback and strategies for autonomous management of time and attention. By notifying students in real time about their distractions, Torddis offers an opportunity to adjust behavior autonomously, thus strengthening metacognitive skills essential for contemporary learning (Ackermans et al., 2025; Al-Gburi et al., 2023).

Additionally, Torddis incorporates principles of adaptive learning by personalizing stimuli according to the learner's characteristics, as suggested by recent studies (Dhananjaya, Goudar, Kulkarni, Rathod, & Hukkeri, 2024; Mwambe, 2022; Sundari, Penthala, & Nayyar, 2024). This personalization, together with the contextualization of learning in real-world environments—aligned with Brown's situated learning model (Brown, Collins, & Duguid, 1989)—makes the system a valuable pedagogical ally in promoting both self-regulation and self-reflection.

2.2 Impact on Cognitive and Socioemotional Development

Students' cognitive and socioemotional development is deeply influenced by their capacity to actively and reflectively engage in learning processes (Mohamed & Lamia, 2018; Palioura & Sapounidis, 2025; Sáez-López, Román-González, & Vázquez-Cano, 2016). Torddis contributes to this development by providing an interactive environment in which students can connect their actions with concrete outcomes through tutor intervention. This fosters awareness of behavior patterns and their impact on academic performance.

Strategies such as collaborative playful environments or gamification, as documented by Echeverría et al. (2011), are consistent with the Torddis philosophy, promoting emotionally engaged and active learning. This approach is essential for equipping students with competences that extend beyond cognition, preparing them to face complex educational challenges with a holistic perspective.

Compared to other systems that operate in controlled contexts—as reported by Ackermans et al. (2025), Berrezueta-Guzman et al. (2021), and C.-H. Wang, Salisbury-Glennon, Dai, Lee, and Dong (2022)—Torddis distinguishes itself by its adaptability to unsupervised settings such as the home. This flexibility fosters a more natural interaction between students and tutors, supporting the development of socioemotional

skills in everyday scenarios. Thus, students learn to self-regulate their behavior not only to enhance academic performance but also to promote emotional well-being and personal motivation [reference anonymised].

2.3 Torddis and Its Contribution to Autonomous Learning

Torddis is specifically designed to reinforce autonomous learning at home. Its architecture enables the monitoring of student concentration and the delivery of immediate feedback to help identify personal obstacles in the learning process (Li et al., 2024). This dynamic supports sustained attention and time self-management, while also fostering intrinsic motivation.

The system's functionality—based on the analysis of facial expressions and behavioral patterns—enables students to adopt personalized improvement strategies. In this way, Torddis not only detects attention-related issues but also actively contributes to the development of self-directed learning awareness, a foundational principle of modern education.

2.4 The Role of Tutors and Parents

Torddis extends its pedagogical impact by facilitating the active involvement of tutors and parents through the provision of historical data on students' attentional behavior. This type of information supports timely interventions aimed at improving self-regulation and optimizing the home learning environment (Loh et al., 2025).

The system enables pedagogical decisions to be based on evidence, enhancing supervision efforts and reinforcing the role of family support as a key component of the educational ecosystem. The visualization of distraction patterns can also aid in understanding factors such as lack of motivation or environmental conditions, allowing tutors to tailor personalized strategies (Ackermans et al., 2025; Navarro et al., 2024).

2.5 Torddis and Digital Competence Development

In the digital age, digital competencies are essential for students to interact effectively with technological tools, maximize learning opportunities, and acquire future-critical skills. Torddis plays a key role in this development by providing an interactive and autonomous environment that encourages problem-solving and effective use of educational technology. These skills also prepare students for more complex academic environments (Mohamed & Lamia, 2018).

The implementation of innovative approaches such as the flipped classroom—where students access digital content outside of class time and use in-class time for practical activities—has proven effective in increasing student motivation and engagement. Torddis complements this model by providing continuous supervision and real-time feedback, thereby enhancing self-regulation and self-directed learning—core elements of digital competence (Echeverría et al., 2011). Its design, aligned with the principles of contemporary pedagogy, solidifies its role as a key tool for promoting autonomy and efficacy in learning processes (Coffman & Taber, 2024).

2.6 Complementary Pedagogical Strategies Integrated into Torddis

Torddis integrates and enhances a variety of contemporary pedagogical strategies that have proven effective in strengthening key competences such as self-regulation, self-reflection, and autonomous learning. Although the system does not directly implement these methodologies, its design aligns with them and amplifies their impact through monitoring mechanisms, alerts, and contextualized feedback.

Notable strategies include visual programming languages such as Scratch, which enable students to explore educational scenarios, make decisions, and construct solutions, thereby promoting computational thinking and metacognitive skill development (Sáez-López et al., 2016). Torddis complements such environments by monitoring whether students remain attentive during home-based tasks, detecting signs of distraction or disengagement, such as drowsiness, absence from the study space, or manipulation of irrelevant objects.

Likewise, the flipped classroom model redistributes instructional time toward autonomous activities at home, making Torddis particularly relevant. Through immediate feedback—delivered via notifications, visual and auditory alerts, and reports to tutors or guardians—the system contributes to the development of self-regulated study habits, even in contexts without direct supervision (Mohamed & Lamia, 2018).

Additionally, active learning approaches such as storytelling with educational robots, project-based learning, and problem-based learning promote reflective and personalized student engagement. While Torddis does not replace these approaches, its ability to detect lapses in concentration and issue alerts supports sustained student engagement, aligning with the principles of personalized and autonomous learning (Palioura & Sapounidis, 2025).

Finally, it is important to emphasize that Torddis is not limited to a function of automated control. By offering accurate data on students' attentional and behavioral patterns, it strengthens the tutor's role, enabling more informed and pedagogically meaningful interventions. In this way, the system acts as a mediator between student autonomy and external guidance, promoting self-management as a central axis of academic and personal development.

3 Related Works

The literature on the use of technologies for detecting and correcting student distractions is limited, particularly regarding the integration of IoT technologies for pedagogical purposes. Previous studies have addressed technological applications in child monitoring, yet few have explored how these technologies can be used to provide immediate feedback that promotes learning self-regulation and enhances educational outcomes in non-formal contexts. The state-of-the-art review process was conducted in accordance with the guidelines established by Kitchenham et al. (2007), whose methodology is presented in the diagram in Figure 1.

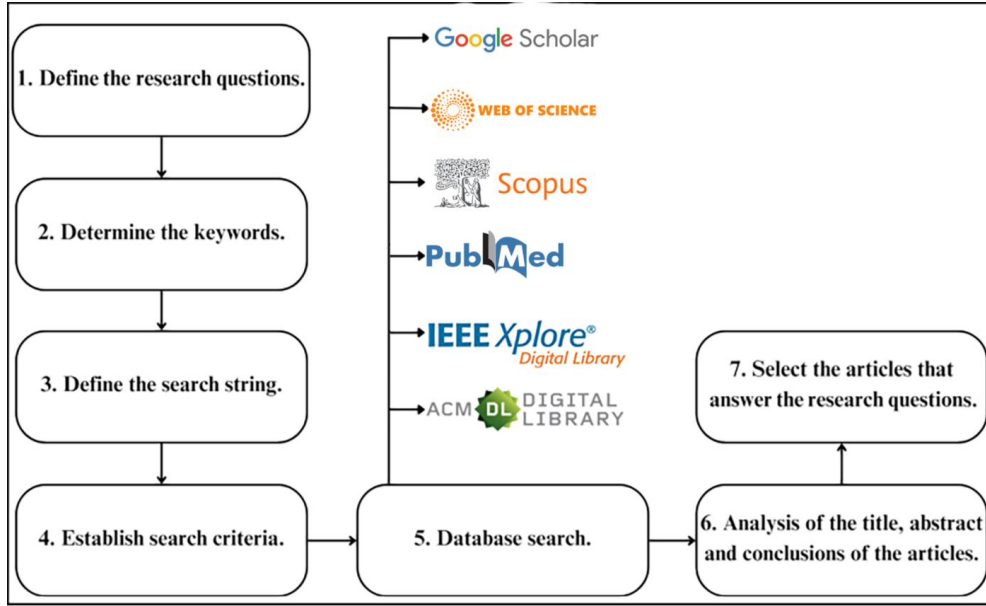


Fig. 1: Literature review scheme.

3.1 Research Questions

This article addresses the critical issue of detecting student distractions during autonomous schoolwork. With the increasing availability of digital tools for education and the growing need for effective learning strategies, understanding how distractions impact student performance has become essential. The following research questions guide this comprehensive review of related works and inform the development of technological solutions aimed at improving children’s concentration and educational outcomes:

- *RQ1*: Which AI and IoT technologies are most suitable for designing a system capable of monitoring children’s distractions during their school activities at home?
- *RQ2*: What internal and external parameters can be used to detect distractions in children, and how should these be integrated into an IoT system to ensure effectiveness and personalization?
- *RQ3*: How can a real-time IoT system foster key skills such as self-regulation, autonomous learning, and concentration in students working on school tasks at home?
- *RQ4*: What are parents’ or tutors’ perceptions regarding the usefulness, usability, and effectiveness of an IoT system designed to support home learning, and how does this perception influence its acceptance?

3.2 Search Strategy

Once the research questions were defined, relevant keywords were selected to retrieve the most recent publications from major scientific databases in this research field. These keywords were used to construct the search string, which was then adapted to the specific requirements of each selected database. The final search string was: (distraction OR distractions) AND (child OR children OR student) AND (homework OR “school assignments” OR “school tasks”) AND (“object recognition” OR “facial recognition” OR “gesture recognition”).

The literature searches were conducted in high-prestige scientific databases, including ACM Digital Library, IEEE Xplore, PubMed, Scopus, and Web of Science. Additionally, Google Scholar was used to expand the search results and minimize the likelihood of omitting relevant studies related to our research. These platforms facilitated the identification of pertinent literature using predefined criteria.

The selected documents came from recognized academic publishers such as ACM, Elsevier, Springer, IEEE, among others, ensuring the scientific quality and validity of the material analyzed. Table 1 presents the results of the initial search in these databases and in Google Scholar. The *Duplicates* column shows the number of documents that had already been retrieved from other databases and were therefore excluded. After analyzing the title and abstract, a full-text review of the documents was conducted to determine whether they addressed any of the study’s research questions. As a result of this final review, the total number of documents included in the analysis is shown in the *Final Results* column of Table 1.

Table 1: Results of the related works search.

Database/Search Engine	Preliminary Results	Duplicates	Title and Abstract Review	Final Results
ACM	1	0	1	1
IEEE Xplore	3	0	2	2
PubMed	2	1	0	0
Scopus	22	3	19	15
Web of Science	8	6	2	2
Google Scholar	15 400	19	16	6

3.3 Literature Analysis

In the scientific literature on technologies applied to the fields of education and child care, several approaches focus on the use of Artificial Intelligence (AI), the Internet of Things (IoT), and computer vision techniques to enhance learning and behavioral monitoring. Additionally, other studies have integrated these technologies to facilitate learning self-regulation, improve concentration, and foster academic engagement across different student populations.

3.3.1 Behavior Monitoring

Studies such as those by [Akter et al. \(2021\)](#), [Albrecht et al. \(2014\)](#), [Berrezueta-Guzman et al. \(2021\)](#), [Pelc, Kornreich, Foisy, and Dan \(2006\)](#), [de Villiers Rader, Zukow-Goldring, and Alhanti \(2021\)](#), [Warren et al. \(2015\)](#), and [Washington et al. \(2016\)](#) explore applications ranging from early diagnosis of Autism Spectrum Disorder (ASD) to improving emotion recognition in children with Attention Deficit Hyperactivity Disorder (ADHD). These studies employ advanced technologies such as AI and connected devices to analyze and respond to specific child behaviors in both controlled and everyday environments.

For example, [Akter et al. \(2021\)](#) apply transfer learning² for ASD diagnosis through facial recognition. Similarly, [Pelc et al. \(2006\)](#) focus on recognizing emotions in children with ADHD using validated photographs, while [Albrecht et al. \(2014\)](#) investigate visual search strategies in children with ASD. [Berrezueta-Guzman et al. \(2021\)](#) evaluate the use of a robotic assistant to support academic tasks for children with ADHD, and [de Villiers Rader et al. \(2021\)](#) examine how gestures can direct attention toward word-object relationships in children with ASD.

Other studies, such as [Boumiza, Bekiarski, Souilem, and Pleshkova \(2017\)](#), [Da Costa Rocha et al. \(2023\)](#), [Enadula et al. \(2021\)](#), [Farsani, Breda, and Sebasti   \(2020\)](#), [Hachad, Sadiq, and Ghanimi \(2020\)](#), [James and Nettikadan \(2019\)](#), [Kulkarni, Anjali, Rohith, Nadagouda, and Veeresh \(2023\)](#), [A.P. Kumar and Kumar \(2024\)](#), [M  ller et al. \(2018\)](#), [Narkhede, Menon, Mathane, Nikam, and Dange \(2023\)](#), and [Ozdamli, Aljarah, Karagozlu, and Ababneh \(2022\)](#), also address issues related to safety, engagement, and interaction through monitoring technologies. For instance, [Hachad et al. \(2020\)](#) and [James and Nettikadan \(2019\)](#) employ facial recognition to monitor student safety on school buses and attendance in classrooms, respectively. Likewise, [Ozdamli et al. \(2022\)](#) use computer vision technologies to recognize emotions and detect cheating during online exams.

In addition, [Campbell \(2015\)](#) and [U  ar and   zdemir \(2022\)](#) present systems designed to monitor distractions in real time. For example, [Campbell \(2015\)](#) use multispectral imaging to detect distractions caused by mobile devices, while [U  ar and   zdemir \(2022\)](#) employ facial recognition and head pose estimation to assess student engagement in virtual classes. Other works such as [Argel, Ragmac, Lindo, and Tiglao \(2023\)](#), [reference anonymised], and [Nguyen, Tieu Binh, Bui, and N.T. \(2019\)](#) also focus on behavioral evaluation using advanced technologies like YOLOv3 for real-time monitoring and dynamic adjustment of teaching strategies.

Finally, studies such as [Boumiza et al. \(2017\)](#) and [Da Costa Rocha et al. \(2023\)](#) explore monitoring in contexts such as automated tutoring systems and school transportation within smart cities, highlighting the use of facial recognition to personalize educational experiences and enhance safety.

² *Transfer learning* [Akter et al. \(2021\)](#) refers to the extent to which learners can effectively and consistently apply the knowledge, skills, and attitudes acquired during a training activity to different contexts beyond the original learning environment. In essence, it assesses whether an educational experience promotes learning that can be generalized and sustained in real-life situations.

3.3.2 Technology for Autonomous Learning

Other studies focus on promoting self-regulation, concentration, and academic engagement by integrating technologies that support autonomous learning in diverse contexts. For instance, [Bembich, Cicognini, and Paoletti \(2016\)](#) investigate the use of digital technologies for future educators, providing self-regulation strategies to manage distractions. Similarly, [Peters \(2003\)](#) examine how hypertext environments can improve volitional control and distraction management in digital autonomous learning.

[Roberts \(2020\)](#) analyze how secondary students use digital records to self-regulate and reflect on their academic progress. On the other hand, [Adcroft \(2018\)](#) combine digital technologies with time management and mindfulness practices to help students improve their focus and self-regulation in daily tasks.

Studies such as [Salter \(2014\)](#) and [Palmer et al. \(2022\)](#) integrate technologies and platforms to monitor and assess self-regulation skills. While [Salter \(2014\)](#) adopt a whole-school approach, [Palmer et al. \(2022\)](#) focus on synchronous hybrid teaching in a pharmacotherapy course.

[Muljana \(2022\)](#) and [C.-H. Wang et al. \(2022\)](#) explore specific strategies to promote self-regulation. [Muljana \(2022\)](#) analyze how instructional professionals use social media to develop self-directed learning, while [C.-H. Wang et al. \(2022\)](#) investigate the reduction of digital distractions among university students through self-regulation skills.

Additionally, [Hartley, Bendixen, Shreve, and Gianoutsos \(2022\)](#) and [Labăr and Țepordei \(2019\)](#) examine the impact of mobile device use on attention and multi-tasking. While [Labăr and Țepordei \(2019\)](#) take a psychological approach to analyze time perception, [Hartley et al. \(2022\)](#) focus on how smartphone use directly affects self-regulation abilities.

These studies reflect a diverse range of approaches to self-regulation and the improvement of autonomous learning, employing technological strategies to reduce distractions, enhance volitional control, and foster academic engagement.

3.3.3 Comparison of Torddis with Previous Works

The literature analysis reveals significant progress in the use of advanced technologies to address educational and child care challenges. However, Torddis stands out as a comprehensive solution that integrates IoT, computer vision, and AI to deliver adaptive and proactive supervision in various environments. The following comparison outlines how Torddis overcomes limitations identified in the reviewed references.

In the domain of behavioral monitoring, works such as [Akter et al. \(2021\)](#), [Pelc et al. \(2006\)](#), and [Albrecht et al. \(2014\)](#) employ facial recognition and visual analysis for diagnosing ASD and ADHD. While these approaches are valuable for understanding behavioral patterns, they lack real-time intervention capabilities—something Torddis provides through proactive alerts that redirect students’ attention. In contrast, studies like [Campbell \(2015\)](#), [Uçar and Özdemir \(2022\)](#), and [Argel et al. \(2023\)](#) focus on identifying distractions but do not include features for adapting to domestic environments, a less structured context that Torddis effectively addresses.

Additionally, research by [Berrezueta-Guzman et al. \(2021\)](#), [de Villiers Rader et al. \(2021\)](#), and [Washington et al. \(2016\)](#) introduces wearable and robotic technologies for children with developmental disorders. While these proposals offer personalized classroom support, they do not extend to home settings or enable continuous monitoring without constant adult intervention. Meanwhile, [Müller et al. \(2018\)](#) focuses on playful activities using facial recognition for entertainment, and [Nguyen et al. \(2019\)](#) employs posture and gesture recognition to assess student engagement in classrooms. In contrast, Torddis balances monitoring of school task performance with strategies to enhance attention.

In terms of safety and educational personalization, studies such as [Hachad et al. \(2020\)](#), [James and Nettikadan \(2019\)](#), and [Boumiza et al. \(2017\)](#) apply facial recognition for administrative tasks such as attendance and school transportation safety. However, they do not address pedagogical integration or adaptive student behavior monitoring. Similarly, systems like those proposed by [Da Costa Rocha et al. \(2023\)](#), [A.P. Kumar and Kumar \(2024\)](#), and [Narkhede et al. \(2023\)](#) employ technologies to detect emotions and attention during academic activities, but they fail to integrate distractor detection and proactive responses as Torddis does.

Regarding technologies for autonomous learning, studies such as [Bembich et al. \(2016\)](#), [Roberts \(2020\)](#), and [Peters \(2003\)](#) focus on promoting self-regulation through digital tools. [Muljana \(2022\)](#) analyze how instructional designers use social media to support self-regulation, but these strategies depend entirely on user commitment and lack automated attention redirection capabilities. In contrast, [Palmer et al. \(2022\)](#), [Adcroft \(2018\)](#), and [Salter \(2014\)](#) introduce pedagogical practices such as mindfulness and hybrid teaching but do not incorporate advanced technologies like those implemented in Torddis for direct learning process intervention.

In the context of multitasking and digital distractions, studies such as [Hartley et al. \(2022\)](#), [Labār and Tēpordei \(2019\)](#), and [Farsani et al. \(2020\)](#) examine the impact of mobile devices on attention and academic engagement. While these works identify key issues, Torddis surpasses them by automatically detecting device-generated distractions and applying immediate corrective strategies. Furthermore, studies like [reference anonymised] and [C.-H. Wang et al. \(2022\)](#) address digital distractions from gestural and pedagogical perspectives, respectively, but do not offer real-time automated interventions.

Finally, works such as [Kulkarni et al. \(2023\)](#), [Enadula et al. \(2021\)](#), [Ozdamli et al. \(2022\)](#), and [Warren et al. \(2015\)](#) highlight the use of advanced technologies for educational interaction and emotion detection. However, these studies do not consider the multidimensional approach that Torddis introduces by combining emotional, cognitive, and contextual analysis to intervene effectively in home environments during academic activities.

In conclusion, while the reviewed studies offer meaningful advances in specific areas, Torddis emerges as an integrated solution that not only overcomes the technical limitations of existing systems but also addresses the dynamic needs of educational and child supervision environments with a proactive and adaptive approach.

4 Torddis Development

The development of the Torddis system followed a structured methodology to ensure the effective integration of hardware and software components, aiming to provide a comprehensive tool for monitoring and improving students’ concentration at home.

More specifically, Torddis was developed using the TDDM4IoTS methodology [reference anonymised]. This methodology is one of the most comprehensive in terms of the system lifecycle [reference anonymised], and has been validated and evaluated in various application domains [reference anonymised]. During the development of the proposed system, all stages of the methodology were applied, except the final one—maintenance—due to the system’s short lifespan at the time of writing this article. Furthermore, this methodological framework facilitates early error detection, contributing to reduced costs and development time while enabling cleaner code (Beck, 2002).

The project team was composed of the authors of this work, with the roles of each member detailed in Table 2.

Table 2: Roles performed by team members.

Role	Member
Client/End User	Author 5, Author 6, and some parents
Project Facilitator	Author 1 and Author 4
Development Team	Author 2 and Author 3
Advisor	Alternating between Author 2 and Author 3

4.1 Preliminary Analysis

This is a critical stage, as it determines whether the development of the project is feasible. To assess feasibility, the activities outlined in the following sections were conducted.

4.1.1 Preliminary Requirements Analysis

This analysis covered both functional aspects (client specifications) and non-functional aspects (system quality attributes) at a general level of detail.

The project team was able to recruit a tutor user with an urgent need for a system like Torddis, enabling a rapid preliminary analysis of their needs. Additionally, a group of tutor users was recruited to verify the proposed requirements in this activity. All these actors later collaborated in performing a detailed analysis of the system requirements.

In this activity, the functional requirements that Torddis must meet were determined (Asish, Kulshreshth, & Borst, 2022; Pabba & Kumar, 2022; Vettivel, Jeyaratnam, Ravindran, Sumathipala, & Amarakehrthi, 2018):

- Register and modify tutor user data.
- Allow the tutor user to register and modify the child's data in the mobile application.
- Allow the tutor user to enable and/or disable permitted objects in the study area (desk).
- Monitor the child's presence, facial expressions, drowsiness state, and the use of non-permitted objects (distractors) in the study area (desk).
- Display a monitoring history of the child's distraction parameters.
- Play an alarm sound and activate an LED light when the child leaves the study area or shows signs of drowsiness.
- Generate statistical graphs of the events detected: drowsiness states, use of restricted objects, and various facial expressions over the past 7 days from a selected date.

The non-functional requirements that the proposed system must meet are:

- The mobile application must have a minimalist and intuitive interface.
- The device must be similar in size to an everyday object familiar to the child (e.g., a toy).
- As it is intended to be placed in a fixed location (designated for schoolwork), the device must be powered by the public electricity system.
- Personal and authentication data must be encrypted to ensure data privacy.
- Use the available home WiFi network.

Use case diagrams were employed for the requirement analysis, using the free web tool TDDT4IoTS available at [URL anonymised for review] and [URL anonymised for review].

The technologies for Torddis development were also determined, including hardware resources and software technologies and tools for mobile application development and hardware configuration. The selection of resources was based on criteria defined by the authors, such as the use of open-source software tools with adequate documentation, meeting the required functionalities, and with which the development team had experience.

Among the advanced AI technologies considered for selection were methods for:

- Facial recognition
- Facial expression recognition
- Sleep detection
- Object recognition
- Monitoring children's distractions according to defined parameters

It was also deemed important that the IoT hardware used in the system meet the required functionalities, be cost-effective, be available in the local market, and be supported by appropriate documentation. Experience of the development team with the chosen hardware was also a key factor.

These technologies allow for real-time monitoring strategies to be tailored to the detected behavior of students, maximizing system effectiveness by providing personalized supervision. Recent research highlights that AI-based tools, such as ChatGPT,

can not only generate adaptive educational content but also deliver personalized feedback. This approach emphasizes the potential of AI to optimize educational processes and foster student-centered learning environments where personalization and supervision are effectively integrated (Li et al., 2024; X. Wang, Maeda, & Chang, 2025).

To determine which distraction parameters may affect children’s concentration during school activities, a literature review was conducted. In addition, user feedback and input from an educational professional were considered.

4.1.2 Feasibility Analysis

This analysis was conducted considering the three aspects defined in the TDDM4IoTS methodology: technical, financial, and operational feasibility. The availability of necessary hardware and software resources was essential to ensure the project’s success. Moreover, the presence of a development team with adequate skills and knowledge enabled the completion of Torddis within a limited budget and a tight schedule.

Technical Feasibility

A systematic analysis of available technologies for developing Torddis was carried out, encompassing hardware components, AI algorithms and methods, and software tools for monitoring distractions in educational settings. This review, guided by the research questions, allowed for the identification and selection of the most suitable technologies to meet the project’s goals. As a result, the system was developed using programming languages and AI algorithms that effectively addressed the defined needs.

The selection of hardware components was based on the following general criteria:

1. **Availability:** The component had to be in stock. Since there is no direct local market at the study location, reliable online stores (such as www.mercadolibre.com.ec) were sought.
2. **Delivery time:** Delivery should not exceed 48 hours.
3. **Cost:** The component had to be affordable. Among reliable vendors, the store offering the lowest price was selected.
4. **Documentation:** Online documentation had to be available to support proper use of the component.
5. **Experience:** The development team had to have experience configuring the component.

The selection of the software technologies used in developing Torddis was based on a set of methodological criteria defined during the planning phase. These criteria ensured that the selected tools addressed both the system’s technical requirements and the specific needs of the educational context. The main factors considered included:

1. **Compatibility with the development environment:** Tools were chosen to ensure interoperability between components, particularly for web services, databases, and AI logic.

2. **Availability of specialized libraries:** Development environments were selected based on the availability of well-established, regularly updated libraries widely used in the scientific and technological communities.
3. **Technological maturity and robustness:** Solutions selected had to be validated in real-world projects, known for their stability, extensive documentation, and scalability.
4. **Development efficiency and error reduction:** Tools and frameworks that offered integrated features for validation, debugging, and security management were prioritized to accelerate development and minimize common errors.
5. **Alignment with the development team’s skills:** Technologies were chosen according to the team’s prior experience to reduce the learning curve and optimize implementation time.
6. **Applicability in educational and mobile contexts:** Operational conditions of the educational environment were considered, including mobile device access, unstable networks, and ease of use for tutors and students.
7. **Free licensing and cost-free use:** Tools with open-source licenses or those free for academic or research use (e.g., MIT, Apache, GPL) were prioritized to ensure sustainability, avoid additional costs, and support replicability in low-resource educational contexts.

Additionally, the development team ensured that the schedule for the Torddis prototype development was within the established timeframe.

Financial Feasibility

In terms of cost, the Torddis prototype is considered highly affordable, as it was developed using free software, AI technologies, and tools. Financial resources were allocated exclusively to the purchase of necessary hardware components.

Operational Feasibility

Given the non-functional requirements of Torddis, the system ensures that tutor users can operate the device and use the mobile application as needed. Therefore, operational feasibility is guaranteed. Moreover, these requirements also ensure the privacy of personal data.

4.2 Technological Layer Design

The goal of this phase is the design of the system architecture. The architecture serves as a key guide for developers to achieve the project objectives with quality (C. Liu, Zhu, Holroyd, & Seng, 2011). To assist tutors, who often have multiple responsibilities inside or outside the home, the construction of an IoT device was considered to monitor the child while performing school tasks.

For the design of Torddis’ architecture, given that the tutor cannot always be present beside the child during homework, the system was conceived to substitute the tutor’s presence by enabling student monitoring during academic activities at home.

In cases where the tutor is away from home, the architecture also considered how to inform them of the events involving the student and how they could remotely encourage the child to continue doing schoolwork.

Analyzing these aspects was crucial for designing a system that supports tutors in guiding their children’s learning and improving their academic performance, particularly by enhancing concentration during schoolwork at home.

4.3 Detailed Requirements Analysis

The requirements analysis was carried out iteratively, that is, for each of the deliverables. The two main deliverables were the IoT device and the mobile application. Furthermore, each of the components into which these deliverables were subdivided was implemented as individual tasks. These tasks spanned from two to six weeks, depending on time availability.

Since the components were developed separately and sequentially, they were integrated progressively with those previously developed. Thus, upon completion of the last component, the deliverable was ready for use and testing.

To detail the requirements, the process began with a literature review and was further refined based on specifications provided by future users. These requirements were documented using semi-structured use cases, adhering to a predefined structure to maintain standardization (Zegzhda, Poltavtseva, Pechenkin, Lavrova, & Zaitseva, 2018).

4.4 Model Generation and Refinement

As TDDM4IoTS [reference anonymised] suggests that not all stages must necessarily be executed, nor in a fixed order, the model refinement stage was not applied.

This stage is typically essential when requirements are unclear at the project’s outset and become more defined as the development lifecycle progresses. However, in the development of Torddis, the requirements were clearly established from the beginning and remained unchanged.

Furthermore, the omission of this stage is also attributable to the tool used—TDDT4IoTS [reference anonymised]—since model refinement is inherently handled within the use cases, with the tool automatically updating the models. Given that the model was generated directly from use case specifications and that Torddis did not involve a complexity level demanding extraordinary modeling efforts, the model produced by TDDT4IoTS was deemed optimal.

The facial and object recognition methods implemented in the system demonstrated efficient response times and high success rates under controlled conditions. These results confirm that the system facilitates the identification of emotions and distractions, optimizing tutor intervention. This positive impact is evident in the system’s ability to reduce the need for constant supervision and improve attention and concentration during students’ schoolwork.

4.5 Test Generation

According to [Sciarra and Elfner \(2024\)](#), the tests required during system development are categorized as follows, and were applied to ensure that Torddis meets all previously defined requirements:

- *Unit tests*, which are the most numerous and simplest. They allow developers to save time by detecting and correcting errors within each software code unit ([Mafi & Mirian-Hosseiniabadi, 2024](#)).
- *Integration tests*, executed during the integration of components into the deliverables, ensure that interacting parts function correctly—for instance, web services with the database ([Späth & Gutierrez, 2025](#)).
- *System tests*, executed when integrating all deliverables into the complete system, to verify its proper functionality ([Smytsek, Eberlein, Serçe, Grunske, & Zeller, 2024](#); [S. Wang, Lian, Li, Marinov, & Xu, 2024](#)).
- *Acceptance tests*, conducted to evaluate the system from the end-user perspective ([Augusto, Morán, Bertolino, de la Riva, & Tuya, 2025](#)).

4.6 Software Generation and Refinement

The software development for the Torddis system was conducted in two main stages: generation and refinement. Initially, the source code was automatically generated by the TDDT4IoT tool [reference anonymised], based on models and tests also automatically generated from the use cases that contained the system specifications.

Subsequently, this initial code was extended by developers to incorporate functionalities that could not be generated automatically by the tool.

During this phase, all software components were verified by executing the pre-designed tests to ensure functionality and alignment with the established requirements.

In the refinement phase, the software was improved to enhance quality and performance. Enhancements included the reduction of temporary variables through the adoption of green computing practices ([Firmansyah, Sudirman, & Putra, 2024](#)), better naming conventions for attributes, methods, and classes to facilitate code readability, and the addition of comments to critical code sections. These improvements resulted in clean, maintainable, and higher-quality code, reinforcing the robustness of the Torddis system as a whole ([Marabesi, García-Holgado, & García-Peñalvo, 2024](#)).

4.7 Hardware and Software Deployment

Each component of the deliverables was deployed in its designated final environment, replicating real-world conditions of the complete system. Components of the mobile application were initially tested using the Android Studio emulator. Subsequently, after generating the APK file iteratively, the mobile app was deployed on a smartphone meeting the system’s minimum requirements.

The components of the final IoT device (Torddis system) were iteratively and incrementally deployed in simulated or testing environments during development phases.

Later, for integration and system tests, the system was deployed in an ideal real-world environment where any student could perform schoolwork.

Functional hardware and software tests validated system performance under optimal conditions, ensuring proper integration among technological components. Nevertheless, additional testing was deemed necessary in adverse scenarios such as environments with low connectivity or high ambient noise. These situations will allow evaluation of the system’s robustness and expand its applicability to more challenging real-life contexts (Falzetti & Cardone, 2024).

4.8 Deliverable Evaluation

Throughout the project, to evaluate Torddis’ functionality and usability, tutors’ capabilities were considered to ensure that the system could remain operational after deployment without requiring ongoing support from specialized personnel. Only in cases of corrective maintenance would individuals with technical knowledge of Torddis be required.

To evaluate system usability, two SUS questionnaires (Brooke, 1996) were applied: one based on Likert-scale statements (see Appendix B.1), and another with open-ended questions (see Appendix B.2), both answered by the students’ tutors participating in the study. Prior to this, all participants signed an informed consent document and completed a demographic survey (see Appendix A) to contextualize each system evaluator.

In addition, the system’s effectiveness was assessed through a seven-item Likert-scale questionnaire (from 1 “strongly disagree” to 5 “strongly agree”; see Appendix B.3). These items addressed: student acceptance of the device, accuracy of event logging, perceived intrusiveness, sustained use during tasks, effects on self-regulation and concentration, academic benefit expectations, and tutor willingness to continue usage. These dimensions enabled a comprehensive evaluation of both the technical functionality and educational impact of the system.

4.9 Maintenance

Maintenance planning must consider the lifecycle of the IoT device components, complemented by preventive maintenance every three months. It is essential to ensure proper environmental conditions, especially cleanliness to minimize dust, and to avoid humidity and direct sunlight.

The frequency of preventive maintenance should match the average lifespan of the least durable component. Furthermore, it is advisable to have replacement parts available and to maintain an up-to-date backup of the software used to configure the IoT hardware.

The mobile application must be maintained at least once a month to manage the increasing volume of stored data, which could otherwise lead to unnecessary resource consumption. In case of changes to the business logic or user requirements—such as modifications to generated reports—a software engineer will be required to implement such updates properly.

System analysis confirms the durability and efficiency of the components used. However, to ensure technological sustainability, regular preventive maintenance assessments should be incorporated, and energy-efficient components should be explored. These measures will not only prolong the device’s operational life but also strengthen its positioning as a responsible and sustainable technological solution (Deng, Jiang, Yu, Lu, & Liu, 2024).

5 Methodological Design of the Study and Experimental Validation

This study is based on a methodological approach that seeks to establish a bridge between technological design and pedagogical processes, with the aim of rigorously evaluating the impact of Torddis in real-world usage contexts. This section outlines the validation strategy employed, understood not only as a technical phase but as a key instance for articulating theory, user experience, and empirical evidence. The adopted methodological structure responds to both scientific validity requirements and the need to generate applicable knowledge in dynamic and diverse educational environments.

5.1 Participants and Study Context

The study involved 12 children and their respective guardians, selected through convenience sampling. The children, aged between 7 and 12 years, were enrolled in primary and lower secondary education. The guardians, aged between 25 and 65, had primary and secondary education levels. Participants resided in both urban and rural areas with access to basic services (electricity, drinking water, internet, and mobile telephony).

The selection criteria ensured that households had access to home internet (Wi-Fi or mobile data) and mobile data on the guardian’s smartphone, which were necessary conditions for system deployment. This approach ensures that the participants reflect the profile of potential Torddis users: families with guardians engaged in activities inside and outside the home, responsible for school-aged children requiring supervision.

The representativeness of the sample is considered acceptable given that real users of the system were involved and the defined inclusion criteria were met. Convenience sampling is appropriate for preliminary investigations such as this, allowing the collection of significant data despite limited availability or willingness of families to participate (Di Pietro & Castaño Muñoz, 2025).

5.2 Experimental Process and Instrument Validation

The experimental process was conducted in two phases, which are described in the following subsections.

5.2.1 Controlled Environment

This phase was designed to evaluate the system’s usability and performance. During the evaluation, guardians used the Torddis prototype with their children for 30 minutes

in a previously prepared environment where all system functionalities were verified. To collect user experience data, SUS-type questionnaires and open-ended questions were administered.

5.2.2 Real Environment

This phase aimed to evaluate system effectiveness. It lasted one week, considering that only a single prototype was available. A Likert-scale questionnaire was used to assess guardians' perceptions of the system's utility and effectiveness (see Appendix B.3), including questions on device attractiveness, notification accuracy, and its impact on children's self-regulation and concentration (Ackermans et al., 2025).

Instrument validity was ensured through expert review, confirming the clarity and relevance of the questions for the study's purpose. In addition, system-generated data (e.g., detection of distracting objects and alerts) were manually verified to guarantee accuracy (X. Wang et al., 2025). The data were also analyzed to determine whether the variables followed a normal distribution, thereby guiding the choice of appropriate statistical methods (ANOVA, Kruskal–Wallis) and correlation coefficients (Pearson and Spearman).

5.3 Randomization and Bias Control

No randomization process was implemented due to the nature of the convenience sample. However, the following measures were adopted to minimize bias (Huang et al., 2025):

- Detailed explanation of each questionnaire item before participant completion.
- Assurance of anonymity and confidentiality of responses.
- Questionnaire administration without the presence of researchers to avoid influencing responses.

Nonetheless, potential bias related to the researchers' proximity to participants is acknowledged, which may have influenced the responses (Di Pietro & Castaño Muñoz, 2025).

5.4 Instruments and Statistical Analysis

To evaluate the effectiveness of the Torddis system, validated instruments and statistical tests were used to analyze guardians' perceptions as well as improvements in students' self-regulation and concentration. Collected data included questionnaire responses and system-generated metrics, allowing an integrated analysis combining subjective and objective perspectives.

5.4.1 Questionnaires and Scales

Three main questionnaires were used to assess various aspects of the Torddis system, in addition to the demographic survey:

- A standard **System Usability Scale (SUS)** questionnaire consisting of 10 items (see Appendix B.1) to assess overall system usability from the guardians’ perspective. A specific preprocessing procedure was followed: negatively worded items were reverse scored, and final results were scaled from 0 to 100, in accordance with SUS guidelines (Brown et al., 1989), ensuring score comparability and alignment with international usability benchmarks.
- An open-ended questionnaire (see Appendix B.2), used to identify areas for improvement before finalizing the system, allowing feedback on usability and effectiveness.
- A Likert-type questionnaire with seven statements (see Appendix B.3), designed to measure *guardians’ perceptions of system effectiveness* in the home setting. Statements evaluated aspects such as device attractiveness (*Statement 1*), discretion during use (*Statement 3*), and perceived improvements in self-regulation and concentration after use (*Statement 5*). This instrument was validated by a panel of experts for clarity and relevance.

The integration of these instruments’ data with system-generated metrics enabled a more robust analysis. Collected metrics included:

- Frequency and duration of homework supervision by guardians.
- Number of system alerts issued to redirect student attention.
- Effective device usage time during homework activities.

Data collected by the Torddis system, such as alert frequency and active supervision time, were incorporated into the statistical analysis to support qualitative findings. Additionally, usage pattern analysis reinforced the validity of guardians’ perceptions regarding system effectiveness and usefulness.

This comprehensive statistical approach ensures that the results accurately reflect the impact of the Torddis system on children’s supervision dynamics while completing homework, providing a solid foundation for future improvements and deployments.

5.4.2 Statistical Analysis

Descriptive and analytical statistical analyses were performed to validate the collected data, including:

- **Descriptive statistics:** Measures of central tendency (mean) and dispersion (standard deviation) were calculated for each statement in the Likert scale, revealing general patterns in guardians’ perceptions (Yaska & Nuhu, 2024).
- **Normality tests:** Shapiro–Wilk (de Souza, Toebe, Chuquel Mello, & Chertok Bittencourt, 2023) and Kolmogorov–Smirnov (Lilliefors, 1969) tests were applied to assess the distribution of numerical variables. Results determined whether parametric or non-parametric tests were used.
- **Correlations:** Pearson (Yu & Hutson, 2024a) and Spearman (Yu & Hutson, 2024b) correlation coefficients were computed to explore relationships between questionnaire statements on system effectiveness (Jebb, Ng, & Tay, 2021). For example, *Statement 5* showed significant correlations with statements on device attractiveness (*Statement 1*) and discretion (*Statement 3*).

- **Qualitative analysis:** Open-ended responses and SUS Likert responses were analyzed alongside system metrics to identify cases where guardians' qualitative perceptions complemented or contradicted quantitative data.
- **Analysis of variance (ANOVA):** ANOVA was applied (Dubey, Bodade, & Dubey, 2025; Rojas, Nussbaum, & Moreno, 2025) to assess differences in perception based on guardians' demographic characteristics, such as educational level or place of residence (urban/rural), identifying contextual influences on system perception.

5.5 Limitations and Generalization

The study has certain limitations:

1. The sample was limited to a small group selected via convenience sampling, which may not represent other sociodemographic or technological contexts.
2. The real-world phase lasted only one week, limiting the assessment of the system's long-term impact on self-regulation skills.
3. Only one prototype was available, restricting simultaneous use by multiple participants.

These limitations are common in pilot or early-stage studies exploring educational technology impact.

Previous studies have highlighted that the generalizability of results from technological interventions strongly depends on population characteristics and implementation context. For example, a recent meta-analysis on the effects of technology on disadvantaged students emphasizes how resource access and contextual conditions can significantly influence outcomes. It recommends designing future studies with broader and more diverse populations to enhance representativeness (Di Pietro & Castaño Muñoz, 2025).

5.6 Validation of the TDDM4IoTS Methodology

The TDDM4IoTS methodology [reference anonymised] was selected for its focus on IoT projects, enabling iterative hardware–software integration. During Torddis development, this methodology supported:

- Definition of functional and non-functional requirements based on guardian needs.
- Automated test generation, reducing development time and errors.
- Early component validation, ensuring system robustness.

Compared to traditional methodologies, TDDM4IoTS stands out for its adaptability to projects with multiple interdependent deliverables, such as those combining diverse IoT components and mobile applications. Furthermore, it ensures compliance with system lifecycle standards [reference anonymised].

Structured methodologies (such as TDDM4IoTS) are particularly effective in the development of technology-based educational systems, as they efficiently integrate technological and pedagogical components (Børte & Lillejord, 2024; Kirkwood & Price, 2014). According to recent studies on adaptive learning systems, applying clear methodological frameworks not only ensures robust development but also

aligns system features with users’ specific needs. This methodological approach is key to addressing interoperability and personalization challenges in educational contexts (X. Wang et al., 2025).

5.7 Pedagogical Justification

The system design and applied methodology align with key pedagogical goals, such as fostering self-regulation and autonomous learning skills in students. Through immediate feedback and personalization, Torddis helps students manage attention and time, while guardians can supervise and support the learning process (A. Chen et al., 2025).

Moreover, the iterative development stages guided by TDDM4IoTS allowed for the incorporation of pedagogical improvements at each phase, ensuring the system addresses both technological and educational needs (Huang et al., 2025).

6 Results and Discussion

The results obtained indicate that demographic factors such as the age or educational level of the guardians do not have a significant impact on their perception of the system’s effectiveness. This confirms that Torddis is an inclusive and versatile tool, capable of adapting to different user profiles. However, it is noted that greater personalization—based on the specific characteristics of the tutoring context—could further enhance its effectiveness and user acceptance (Conati, Barral, Putnam, & Rieger, 2021).

The following subsections present the key findings derived from each development phase of the system, corresponding to the applied methodology. These findings include results from functional tests, usability questionnaires, and analyses of demographic and monitoring data. These evaluations, conducted on the Torddis prototype, aim to assess both its technical performance and its acceptance from a user experience perspective.

The outcomes obtained enable a comprehensive evaluation of the Torddis system’s performance in domestic settings, providing empirical evidence of its utility as a technological tool for learning support. These results stem from the systematic application of the stages defined in the TDDM4IoTS methodology [reference anonymised], whose guided implementation structured the prototype’s development and facilitated the validation of its pedagogical and technical effectiveness.

6.1 Preliminary Analysis

The coarse-grained analysis of the functional requirements is presented in the general use case diagram shown in Figure 2. These functionalities and their corresponding actors were identified as a result of the initial interviews with the end user (a guardian).

The *guardian* user is responsible for configuring the *allowed objects*, *monitoring* their *students*, receiving alerts issued by the system regarding detected events, and persuading the monitored student to remain focused. Additionally, the guardian can access system-generated reports and review the history of recorded events.

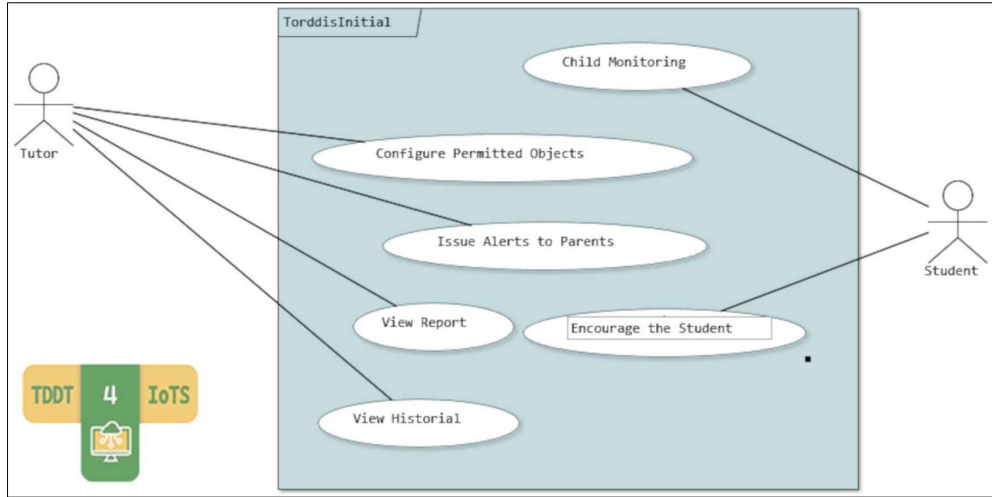


Fig. 2: High-level user requirements modeling.

6.1.1 Hardware Resources

The hardware used for building the Torddis system device was selected based on criteria of integration, low power consumption, and compatibility with real-time monitoring environments. Table 3 presents the main components employed, along with their corresponding functionalities, evaluated alternatives, and technical remarks. The ESP32-CAM and NodeMCU ESP8266 modules stand out due to their wireless connectivity and embedded processing capabilities, making them suitable for video transmission and information exchange with the web service. Furthermore, peripheral components such as passive buzzers and 5 mm LEDs were incorporated to emit auditory and visual alerts, thereby fulfilling the system’s functional requirements for home-based educational supervision.

Table 3: Selected hardware components and technical observations

Functionality	Selected	Alternatives	Technical Observation
Web service connectivity and video streaming	ESP32-CAM	OV7670 VGA camera	This module includes Wi-Fi + Bluetooth connectivity and an integrated video camera (Casas Sánchez, Loayza Apeña, Palomino, & Paiva-Peredo, 2022).
Data transmission	NodeMCU ESP8266 v3	Arduino UNO, GSM module	The NodeMCU is a compact board that provides Wi-Fi connectivity for data transmission to the web service (Barai, Rysul Kibria Badhon, Zhora, & Rahman, 2019).
Alarm sound emitter	Passive buzzer	Active buzzer	Commonly used to generate audible alerts in electronic boards (Adebisi, Adejumobi, Durodola, & Jim, 2023).
Visual alert emitter	5mm LED (any color)	RGB LED	Bright clear LED diode used to provide visual alerts (Upender, Reddy, & Santoshini, 2020).

6.1.2 Software Technologies

The system’s development enabled the effective integration of diverse technologies, including programming languages and AI algorithms, which facilitated the achievement of the proposed objectives and demonstrated its technical feasibility. Table 4 presents the software technologies used in the development of the Torddis system.

Table 4: Software technologies used in the development of the Torddis system

Functionality	Selected	Alternatives	Technical Observation
Android mobile application	Java	Kotlin	Language used in Android Studio, with a large developer community in the mobile application domain (Sharma, Bux, Varshney, & Tomar, 2021).
AI applications	Python	Java, R	Programming language with popular open-source libraries for AI application development (Cai, Langtangen, & Moe, 2005).
Web applications and services	Django	Flask	Popular framework with rapid development capabilities that helps prevent security flaws in web applications and services (Puneet et al., 2022).
Databases	PostgreSQL	MySQL, MariaDB, Firebird	Robust database with enhanced compatibility with the Django web framework (Puneet et al., 2022).

6.1.3 Distraction Parameters

The distraction parameters identified in the literature as factors affecting children’s concentration during academic activities were refined based on user feedback and expert input from an education professional. Table 5 summarizes the distraction parameters that were defined by consensus.

Table 5: Distraction parameters identified in the study area

Internal Distraction	External Distraction
Emotional expressions such as anger, disgust, fear, happiness, sadness, surprise, and neutral state (Asish et al., 2022; Pabba & Kumar, 2022; Vettivel et al., 2018).	Intervention of an unfamiliar person or abandonment of the study area (Vettivel et al., 2018).
Drowsiness state to determine whether the child is awake or asleep (Pabba & Kumar, 2022).	Presence of objects in the study area that divert attention from the ongoing academic tasks (Asish et al., 2022; Pabba & Kumar, 2022).

6.1.4 AI Methods for Monitoring Children’s Distractions

In this stage, the most suitable AI methods were selected to implement the Torddis proposal. These include methods for:

- Facial recognition,

- Facial expression recognition,
- Drowsiness detection, and
- Object recognition.

Facial Recognition Methods.

Facial recognition methods focus on face detection by identifying patterns such as eyes, lips, mouth, and nose, among other facial features. Table 6 lists several methods used for face detection and recognition.

Table 6: Facial recognition methods

Method	Implementation Environment	Purpose	Accuracy
HCC ^a	OpenCV	Face detection in low-light environments (Le & Mohd, 2022).	81.00%
Amazon Rekognition	(not specified) AWS ^b	Gender classification (Goel & Agarwal, 2012). Authentication through facial recognition (Girmay, Samsom, & Khattak, 2021).	98.75% 100%
YOLO ^c , MTCNN ^d , and FaceNet with SVC ^e	Google Colab	Attendance registration based on deep learning (Darapaneni et al., 2020).	99.00%
LBPH ^f	(not specified)	Facial detection from captured images (Garcia, Lacayanga, & Cruz, 2021).	91.72%

^aHaar Cascade Classifier: A fast object detection algorithm that uses Haar-like features to detect faces in images. ^bAmazon Web Services. ^cYou Only Look Once: A real-time object detection algorithm. ^dMulti-task Cascaded Convolutional Neural Network: A face detection algorithm.

^eSupport Vector Classifier: A supervised learning technique commonly used for data classification.

^fLocal Binary Patterns Histogram: A facial recognition algorithm that uses local binary patterns and compares histograms for face identification.

The combination of YOLO with MTCNN and FaceNet with SVC was deemed unsuitable for the proposed system due to its computational complexity and high resource requirements. Additionally, Amazon Rekognition was ruled out as it is a commercial service with a significant cost. Therefore, the selected approach consisted of the *HCC* method along with the *LBPH* algorithm from the *OpenCV* library, as it enables easy customization and offers relatively short recognition times compared to other alternatives.

Facial Expression Recognition Methods.

Facial expression recognition requires an input image from which features are extracted and subsequently compared against a computational model. Table 7 lists several methods used for facial expression detection.

As indicated in Table 7, the CNN (Convolutional Neural Networks) method was selected for the development of Torddis due to its accuracy. However, it is essential to ensure proper training of the model. When selecting the most suitable method or algorithm for implementation in a system, it is important to consider that the accuracy of certain algorithms depends on the dataset used.

Table 7: Facial expression recognition methods

Method	Dataset Used	Tools and Frameworks	Purpose	Accuracy
HCC	Custom	<i>(not specified)</i>	Recognizes seven expressions: happy, sad, angry, scared, disgusted, surprised, and neutral (Lalitha, Aishwarya, Shivakumar, Srilekha, & Kartheek, 2021).	78.00%
CNN	FERC ^a 2013	Python, Keras, TensorFlow, and OpenCV	Recognition of basic human emotions (anger, fear, neutral, happiness, sadness, surprise, etc.) (Kedari, Kapile, Kadole, & Jaikar, 2021).	60.00%
		<i>not specified</i>	Emotion recognition through facial expressions (Jaiswal, Krishnama Raju, & Deb, 2020).	70.14%
		<i>(not specified)</i>	FER ^b -based classification using static images (Singh & Nasoz, 2020).	61.70%
	JAFFE ^c	<i>(not specified)</i>	Detecting emotions through facial expressions (Jaiswal et al., 2020).	97.97%
	CK+ ^d	Python, Keras, TensorFlow, and OpenCV	Recognition of basic human emotions such as: anger, fear, neutral, happiness, sadness, and surprise (Kedari et al., 2021).	99.10%
	CK+	Python, 2-layer Sequential CNN, Leaky ReLU, Softmax, Dropout	Facial emotion detection and generation of motivational quotes for each emotion (Bikku et al., 2022).	85.70%
	<i>(not specified)</i>	CNN with 80 epochs ^e	Automatically generates and categorizes questions, evaluates answers, and tracks performance by providing motivational quotes when detecting student emotions (Silva, Sudasinghe, Hansika, Gamage, & Gamage, 2021).	99.00%
MobileNetV2	<i>(not specified)</i>	<i>(not specified)</i>	Detects a person's emotion through facial expression using an artificial neural network (M. Kumar & Srivastava, 2021).	89.91%
			Online evaluation of students' ability to train and deploy deep learning solutions (Ilić et al., 2021).	60.00%

^aFacial Expression Recognition Challenge. ^bFacial Expression Recognition. ^cJapanese Female Facial Expression dataset. ^dCK+ refers to the Extended Cohn-Kanade Dataset. ^eAn epoch refers to one complete pass of the model through the entire training dataset, during which each sample is used once to update the model weights.

Facial Expression Recognition Procedure.

The procedure for facial expression recognition is schematized in Figure 3. The process begins with the capture of an image containing the user's face. Subsequently, facial features are extracted using computer vision techniques, which are then compared with a previously registered set of images to identify the user.

The computational model employed was trained on a dataset representing the seven universal facial expressions: anger, disgust, fear, happiness, sadness, surprise, and neutrality, thus enabling accurate emotion classification based on visual patterns (Zhang & Yu, 2022).

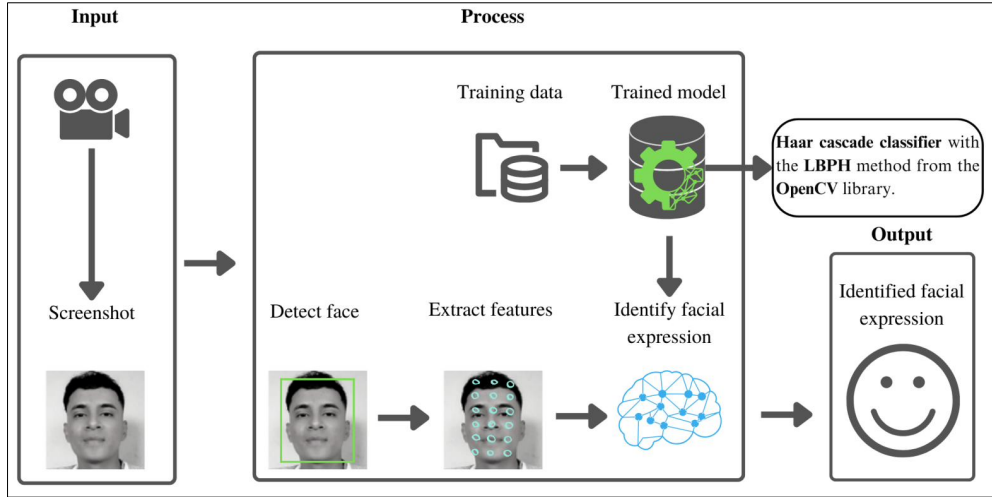


Fig. 3: Facial expression recognition procedure used in the Torddis system.

Drowsiness Detection Methods.

Drowsiness detection methods require an input image containing a human face. These methods perform image preprocessing to extract characteristic eye landmarks and then analyze whether the subject’s eyes are closed. Table 8 summarizes the methods used for drowsiness detection.

Table 8: Drowsiness detection methods.

Algorithm	Purpose	Accuracy
MediaPipe Mesh	Face	
SVM	Detection of facial landmarks with 468 3D points (Shanmugam, Badruddin, & Asirvadam, 2022). Eye tracking (Altameem, Kumar, Poonia, Kumar, & Saudagar, 2021).	99.87% 83.25%
Viola-Jones	Drowsiness detection (Teja & Kumar, 2021).	92.00%
CNN	Trained on four gestures: eyes open, eyes closed, yawning, and not yawning (I, Ramli, Azri, Aliff, & Mohammad, 2022).	80.00%

The SVM (Altameem et al., 2021), CNN (I et al., 2022), and Viola-Jones (Teja & Kumar, 2021) algorithms were discarded due to their limitations in accuracy for drowsiness detection and their complexity in real-time implementation. Specifically, SVM and Viola-Jones require intensive preprocessing and struggle to handle variations in lighting and posture, whereas CNNs, although accurate, demand high computational capacity for inference on low-power devices.

For these reasons, the selected algorithm was MediaPipe Face Mesh (Shanmugam et al., 2022), as it offers an optimal balance between accuracy and computational

efficiency, enabling real-time drowsiness detection with low latency and without the need for specialized hardware.

Object Recognition Methods

Computer vision enables the automatic detection of the structure and properties of a potential dynamic three-dimensional world. First, an input image containing one or more objects is required. Then, image preprocessing is performed to extract features from the segmented image. Table 9 describes a list of object recognition algorithms.

Based on the information presented in Table 9, the selected algorithm was *Sequential* from the *TensorFlow* library in combination with *Keras* due to its accuracy, optimal performance, ease of training, and widespread use for real-time object recognition. MobileNetV2 combined with SSD was discarded due to its lower object recognition accuracy compared to other models. YOLO was also discarded because of the high GPU (Graphics Processing Unit) requirements for proper operation.

Object Recognition Procedure

This method consists of the following steps:

1. A photograph is taken to capture the objects on the desk or in the area where the child is working.
2. The image is preprocessed and segmented into parts corresponding to potential objects.
3. Feature extraction is performed on the segmented image.
4. Finally, the extracted features are compared with a computational model previously trained on a dataset containing the objects to be recognized.

Computer vision enables the automatic detection of the structure and properties of a potential dynamic three-dimensional world based on one or more two-dimensional images of the scene (Cruz et al., 2013). Figure 4 illustrates the object recognition procedure used in the Torddis system implementation.

The analysis highlights that Torddis outperforms existing market solutions due to its integration of facial and object recognition technologies to effectively monitor distractions. This advantage is reflected in the high acceptance rates among tutors and the positive results observed in students, positioning the system as a superior solution for monitoring and supporting academic tasks at home (Deng et al., 2024).

6.2 Technological Layer Design

Following the elicitation of system requirements, the architecture of the Torddis system was initially designed (see Figure 5). Subsequently, with the assistance of the TDDT4IoTS tool [reference anonymised], the IoT device of the system was designed.

Figure 6 illustrates the structural design of the IoT device, highlighting the components used in its construction, which are:

1. ESP32-CAM board
2. ESP8266 NodeMCU board
3. Protoboard

Table 9: Algorithms for object recognition.

Algorithm	Technology and Implementation	Purpose	Accuracy
CNN	Python, TensorFlow, Keras, CIFAR-10 (25 training epochs)	Image classification and recognition, specifically of airplanes (Sudharshan & Raj, 2018).	96.00%
Viola Jones	Haar-like features, AdaBoost, Cascade Classifier	Face detection and eye tracking (blinking detection, eye position) (Teja & Kumar, 2021)	Face: 98% Eyes: 94% With glasses: 82% Without glasses: 92%
PERCLOS ^a	EAR ^b computed from IR ^c images, blink monitoring	Determine driver drowsiness level through eyelid closure percentage (Teja & Kumar, 2021)	Not applicable (dependent on EAR)
YOLOv4 + CNN-29	Python, OpenCV, Darknet YOLOv4	Automatic detection of faults in solar panels (defective cells, hotspots, electrical failures) using thermal images captured by drone (Zou & G V, 2022)	Solar panel: 100% (confidence); Defective cells: 32% initially, up to 89% after retraining
YOLO	CNN (YOLO), bounding boxes and IOU ^d score	Object detection in the environment (e.g., bottles) to trigger alerts (Teja & Kumar, 2021).	98.00%
YOLOv4	Implemented with Darknet and optimized with TensorRT ^e on Jetson Nano platform	Real-time detection of Vietnamese vehicles (Minh, Mai, & Minh, 2021).	90.00%
		Real-time detection of Vietnamese license plates (Minh et al., 2021). Detection of objects and people (F.-Y. Liu, Liao, Chou, & Fan, 2021).	92.00% mAP ⁱ : 73.01%
MobileNetV2-SSD	Implemented with PyTorch and accelerated with TensorRT	Real-time detection of Vietnamese vehicles (Minh et al., 2021).	85.00%
		Real-time detection of Vietnamese license plates (Minh et al., 2021).	90.00%

^a PERcentage of eye CLOSure: metric used in drowsiness detection systems. ^b Eye Aspect Ratio: metric relating eye height and width to estimate openness. ^c Infrared: type of image captured via IR sensors. ^d Intersection over Union: metric measuring overlap between predicted and actual bounding boxes in object detection tasks. ^e TensorRT: NVIDIA inference optimization library that accelerates deep learning models for efficient execution on GPU (Graphics Processing Unit). ^f 2D: bidimensional, two-dimensional matrices in images. ^g MSAM (Modified Spatial Attention Module): spatial attention module that enhances the network's focus on relevant image regions. ^h PASCAL VOC 2007/2012: benchmark datasets for object detection (people, vehicles, animals). ⁱ mean Average Precision.

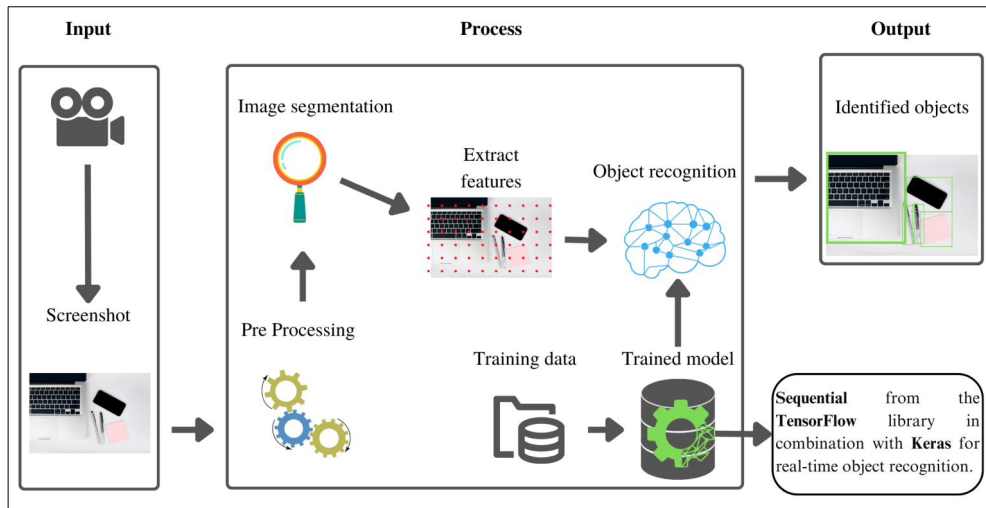


Fig. 4: Object recognition procedure used in the Torddis system.

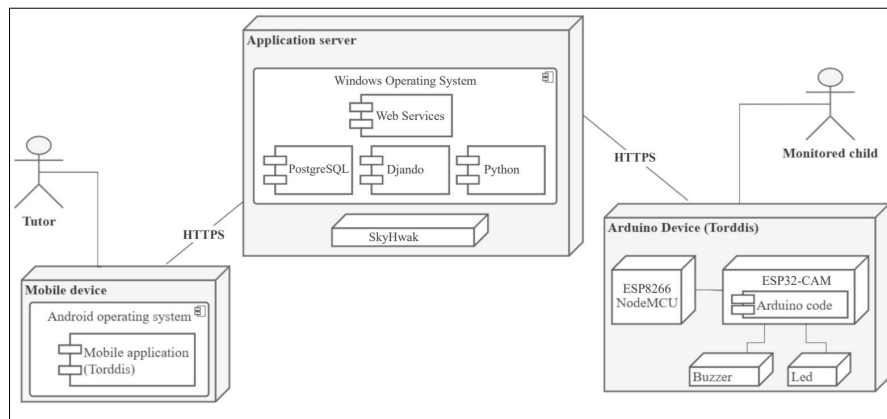


Fig. 5: Deployment diagram (nodes) of the Torddis system.

4. Passive buzzer
5. Red LED

6.3 Detailed Requirements Analysis

With the system architecture and coarse-grained functionalities already defined, the fine-grained requirements were elicited to ensure a successful implementation of Torddis. Figure 7 presents the use case diagram that clearly and specifically represents the problem domain.

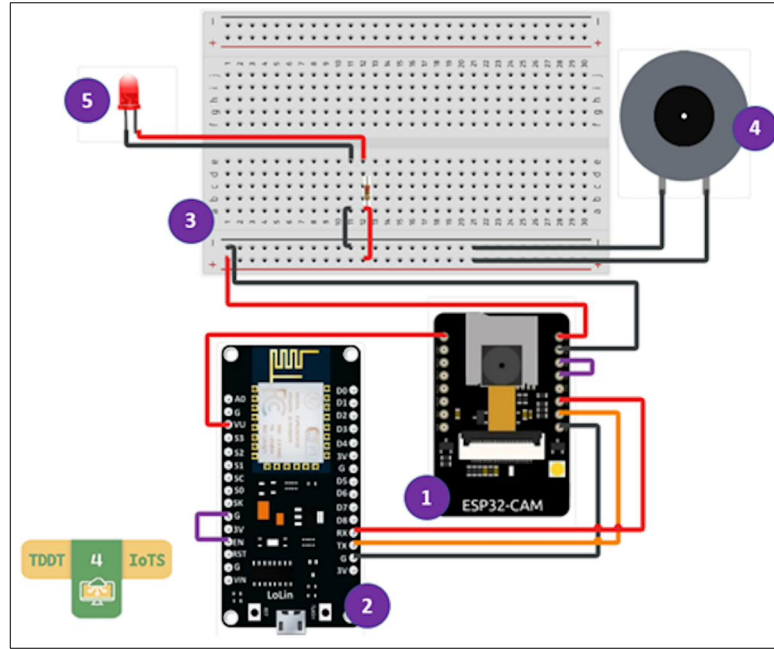


Fig. 6: Design of the IoT device of the Torddis system.

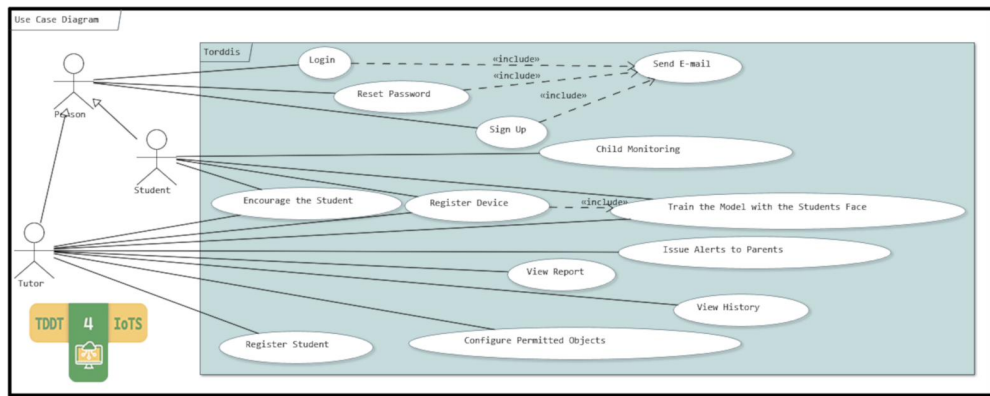


Fig. 7: Fine-grained use case diagram of the Torddis system.

Table 10 presents one of the fine-grained use cases in an extended format so that developers can implement the corresponding business logic and ensure the system meets the established specifications.

Acknowledging technical limitations—such as object detection error rates and resource consumption—has made it possible to identify concrete areas for improvement. The implementation of lighter and optimized models, together with real-time

Table 10: Use Case: Register Torddis device

Use Case	Register Torddis device
Description	The tutor enters the MAC address of the Torddis device. The system registers the device in the database and displays a confirmation message.
Actors	Tutor
Preconditions	User is authenticated as a tutor.
Postconditions	The tutor's Torddis device is linked.
Main Flow	1. The use case begins when the tutor accesses the device-linking interface. 2. The system displays a form with the controls to link the device: MAC address input. 3. The tutor enters the MAC address and selects the save option (see Alt 1). 4. The system stores the data in the database (see E1). 5. The system provides feedback indicating a successful operation. 6. The use case ends when the system displays the facial training interface.
Alternative Flows	Alt 1. The tutor does not wish to register the Torddis device: 3. The use case ends when the tutor cancels the operation.
Exceptions	E1. Error when registering the Torddis device: 4. The system displays a message: <i>Error registering the Torddis device. Please try again.</i> 5. The tutor resumes from step 3 of the main flow.

calibration mechanisms, will ensure greater accuracy and performance of the system, even in less favorable environments (Falzetti & Cardone, 2024).

6.4 Model Generation and Refinement

Among the system modeling tools employed, structural models were used—such as class diagrams (see Figure 8)—which were generated directly by the TDDT4IoTS tool [reference anonymised] based on the use cases that specify the system to be developed.

In addition, the AI computational models for monitoring learner events based on specified distraction parameters were obtained and refined by adjusting configurations such as the number of hidden layers and the number of training epochs.

6.5 Test Generation

The unit and integration tests that can be written in code (such as those related to database connections, web service testing, among others) were also automatically generated by the TDDT4IoTS tool [reference anonymised]. The data for unit

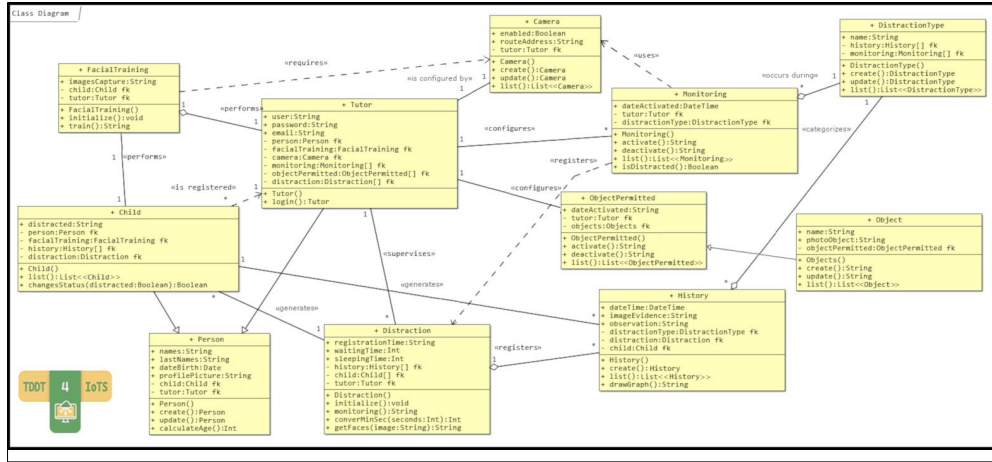


Fig. 8: Class diagram of the Torddis system.

testing (e.g., mobile application functions such as alert playback and drowsiness detection) were mainly obtained during the detailed requirements analysis phase, whereas the integration tests were derived during the system architecture design phase (i.e., communication between the ESP8286 NodeMCU and web services). Some of these integration tests were manually written and executed, such as the interface with the web services.

Figure 9 illustrates the relationship between the test data acquisition and the system life cycle phases. All system and acceptance tests were manually written and executed.

Tables 11, 12, 13, 14, and 15 summarize system test cases that were also used as acceptance tests, enabling verification that Torddis satisfies the defined requirements. These tests were authored and executed during the project—including validation with tutor users—to ensure alignment with the specifications.

Table 11 introduces the test case for tutor account creation, which serves as the system’s entry point. It is part of the documented system and acceptance test suite. The objective is to ensure that tutor registration is correctly recorded in accordance with the specifications (Sciara & Elfner, 2024).

Table 12 describes the test case that verifies signing in with a tutor account, explicitly included among the acceptance tests. It is also aligned with the non-functional requirement mandating encryption of personal and authentication data, thereby strengthening privacy during access.

Table 13 presents the test case for a student’s facial training; it requires a previously registered Torddis device and is included in the acceptance tests. The inclusion of this workflow is consistent with the observed performance of the recognition methods under controlled conditions.

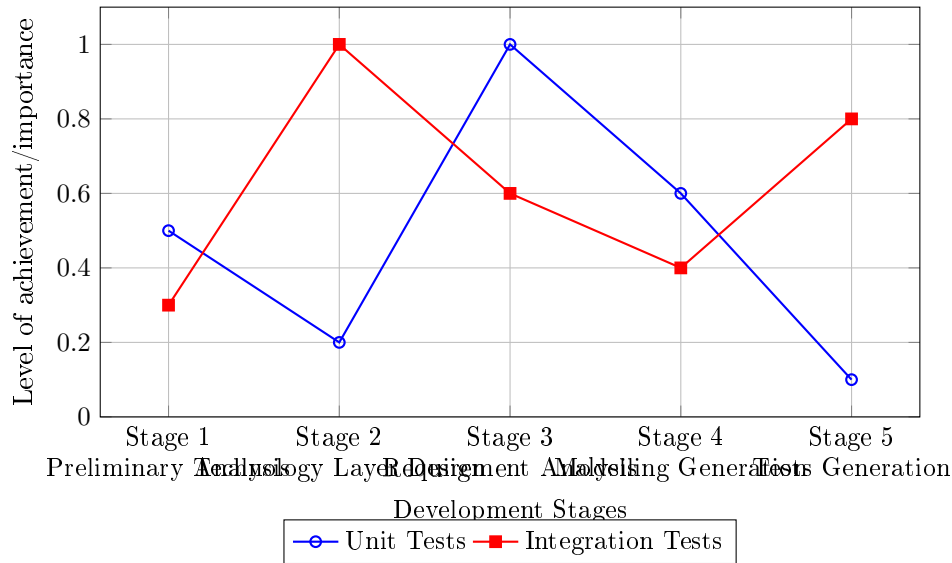


Fig. 9: Relationship between test data Acquisition and the Torddis life cycle phases.

Table 11: Test case for tutor account creation.

Test Case ID:	PU-1
Test Case Name:	Tutor Account Creation
Expected Outcome:	Successfully create a tutor account
Test Case Procedure	
Step No.	Step Description
1	Open the mobile application.
2	Select the “Sign Up” option.
3	Enter the registration data. • First name: Carlos Iván • Last name: Almeida Dueñas • Email: carlos.almeida2017@utec.edu.ec • Date of birth: 10/01/2000 • Username: calmeidad • Password: XXXXXX
4	Select the “Create Account” option.

Table 14 shows the test case that verifies linking the device to the tutor’s account via its MAC address; upon successful completion, the system displays the facial-training interface. The steps confirm input of the MAC address and its persistence in the mobile application.

Table 12: Test case for login.

Test Case ID:	PU-2
Test Case Name:	Login
Expected Outcome:	Successfully log in with a tutor account
Test Case Procedure	
Step No.	Step Description
1	Enter login credentials. • Username: calmeidad • Password: XXXXXX
2	Select the “Login” option.
3	The mobile application displays the tutor menu screen.

Table 13: Test case for facial training.

Test Case ID:	PU-3
Test Case Name:	Facial Training
Expected Outcome:	Successfully perform facial training for a student
Test Case Procedure	
Step No.	Step Description
1	Enter the “Child” module of the application.
2	Select the “Train” option for a child.
3	Register a device (see the test case shown in Table 14).
4	Select the “Train” option.
5	Position the child’s face in front of the device’s camera and wait for the facial training process to complete.

Table 14: Test case for device registration.

Test Case ID:	PU-4
Test Case Name:	Device Registration
Expected Outcome:	Successfully register the device in the mobile application
Test Case Procedure	
Step No.	Step Description
1	Select the “Register Device” option.
2	Enter the MAC address of the device.
3	Select the “Save” option.

Table 15 documents the test case that ensures the tutor can enable or disable permitted objects for distraction monitoring, as specified by the functional requirement, and describes the effective procedure within the application.

Table 15: Test case for configuring object usage.

Test Case ID:	PU-5
Test Case Name:	Object Usage Configuration
Expected Outcome:	Successfully configure object usage
Test Case Procedure	
Step No.	Step Description
1	Enter the “Objects” module of the application.
2	Search for the object to be activated/deactivated.
3	Select the object’s switch control to activate or deactivate it.

6.6 Software Generation and Refinement

Part of the software code was automatically generated by the TDDT4IoTS tool [reference anonymised], and later refined and completed manually by the developers. The development team was responsible for delivering tested and functional software for each deliverable.

To complete the generation of the software code for the mobile application, the Android Studio IDE [Android Studio \(2024\)](#) was used with Java as the programming language, based on the code generated by the TDDT4IoTS tool. In addition, Python was used in conjunction with Visual Studio Code ([Microsoft Corporation, 2024](#)) to implement the web services, and C++ with the Arduino IDE ([Arduino, 2024](#)) to configure the hardware components of the IoT device within the Torddis system.

Consequently, this phase included both unit testing of the individual components of each deliverable and their respective integration tests. In summary, the outcome of this stage was a functional version of the software for each deliverable, thus validating its behavior according to the specified requirements.

6.7 Software and Hardware Deployment

The various components of the mobile application, as well as the complete app, were deployed on a Xiaomi Note 10 Pro device running Android version 13. It is worth noting that the application’s architecture allows deployment on any mobile device running Android version 13 or higher, ensuring broad compatibility within the Android ecosystem.

Figure 10 displays screenshots of both the mobile application and the Torddis device, highlighting its intuitive design and the key features for real-time monitoring of student distractions. The IoT device, shown as well, integrates facial recognition and object detection technologies, providing comprehensive support to improve student concentration during schoolwork at home. Figure 10a shows the different options available to the tutor, while Figure 10b shows the list of students under a tutor’s supervision. On the other hand, Figure 10c presents the Torddis system device, which integrates the camera used to monitor the children while they perform their academic activities.

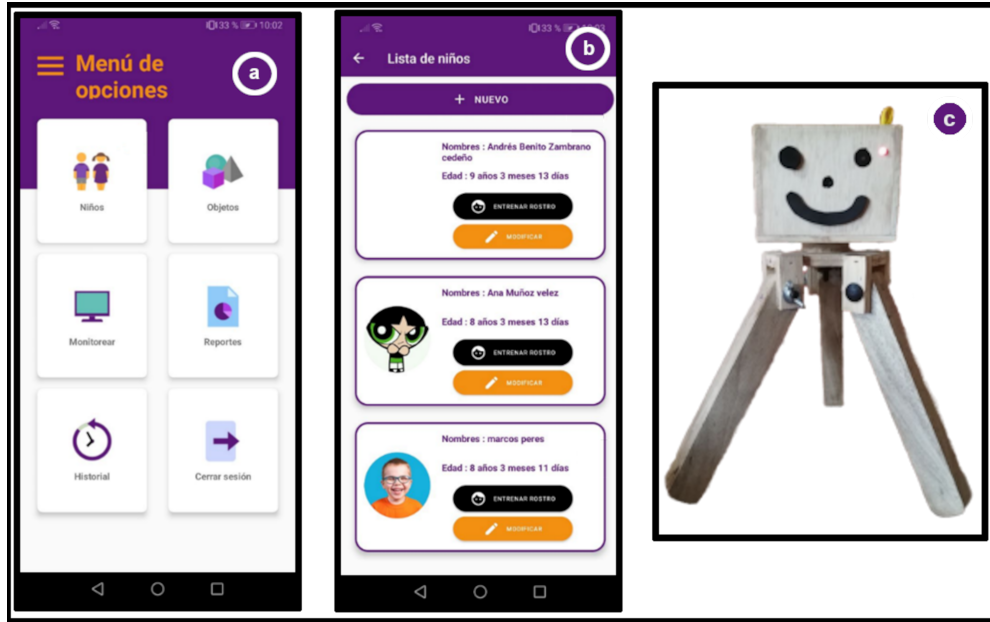


Fig. 10: Screenshots of the mobile application and the Torddis system device.

6.8 Deliverable Evaluation

Each deliverable (mobile application and device) was evaluated by the user in an ideal setting. Figure 11 shows the configuration of one of the environments used to evaluate the Torddis system. The child is seated at a desk performing school tasks while being monitored by the Torddis system.

The system is capable of monitoring the student's concentration and identifying distractions. The device is placed at a distance of 50 to 70 cm from the child to ensure optimal camera operation, which provides the necessary input for facial recognition and object detection technologies. It is worth noting that, from the child's perspective, the device is intended to be perceived as a decorative item on the desk.

An evaluator (one of the authors of this article), along with one or both parents, is present (seated in front of the TV). They are positioned approximately three meters from the child. The evaluator supervises the system's operation via a laptop, while the parent(s) use the mobile application integrated with the IoT device to monitor the child's concentration in real time.

This setup allowed for a comprehensive evaluation of the Torddis system, ensuring that all involved parties (children, parents, and evaluators) could interact with the technology and provide valuable feedback on its effectiveness in monitoring and enhancing concentration during academic activities.

To evaluate the system's functionalities as a whole, acceptance tests were conducted as detailed in Tables 11, 12, 13, 14, and 15.

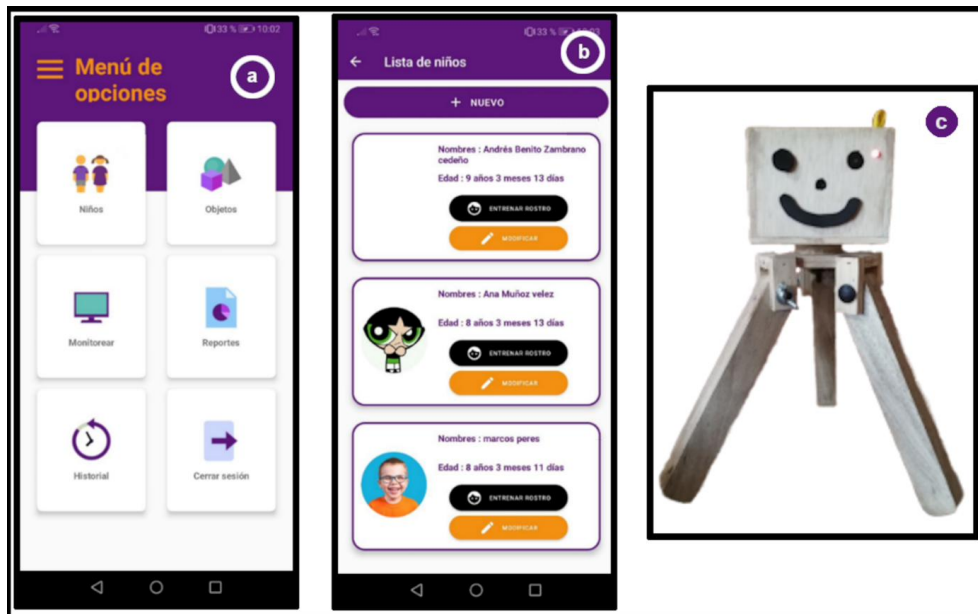


Fig. 11: Setup of the environment for evaluating the Torddis system.

6.8.1 Usability Evaluation of the Developed System Prototype

The usability evaluation of the Torddis system was conducted to collect information about the overall user experience, focusing specifically on ease of use, interface design, and the effectiveness of the features provided. This comprehensive evaluation aimed to identify strengths and areas for improvement, ensuring that the system effectively meets user needs. The following subsections detail the demographic data of the participants, their experiences, the specific tasks performed during the evaluation, and the results obtained through the use of Likert-scale questionnaires (see Appendix B.1) and open-ended questions (see Appendix B.2) from the System Usability Scale (SUS) (Brooke, 1996).

Demographic Data.

At the beginning of the usability evaluation of the Torddis system, a demographic questionnaire was provided (see Appendix A) to gather information about the tutors who participated in the evaluation. The questionnaire yielded the following results:

- A total of 12 families were evaluated, represented by 8 mothers and 4 fathers, all from the geographic area of a Latin American educational setting.
- The average age of the participants was 39 years, ranging from 25 to 65 years.
- 58% of the tutors had completed only primary education, while 42% had reached secondary education.

About Monitoring.

One of the preliminary questions in the usability evaluation focused on how easily the tutors could monitor their children while performing school tasks. Most tutors responded that monitoring for signs of distraction during homework was a complex task, as shown in Figure 12a, which presents the responses to that question. Another question asked about how frequently they needed to monitor their children during such activities. Most tutors indicated that they had to do so constantly (as shown in Figure 12b). Finally, tutors reported that among the strategies they use to help their children stay focused during school activities are scolding, motivational advice, and rewards such as candies, toys, trips to the park, favourite meals, among others, as shown in Figure 12c.

Tasks Performed by Participating Guardians.

The tasks performed by the guardians during the usability evaluation were as follows:

1. Connect the Torddis device to the power source and place it on the desk where the child will sit to complete schoolwork.
2. Create a tutor user account in the mobile application.
3. Log in to the mobile application.
4. Register the child to be monitored.
5. Register the Torddis device in the mobile application using the MAC address provided on the device's label.
6. Perform facial training for the registered child.
7. Enable and/or disable the objects (distractors) to be monitored in the corresponding section of the mobile application.
8. Assign a task to the child while being monitored by the Torddis device: colouring a mandala.
9. Activate the detection of each distraction parameter in the monitoring section of the mobile application.
10. The child remains unsupervised for 6 minutes, in the passive presence of adults (independence condition).
11. Enable and/or disable video streaming from the Torddis device.
12. Browse the history of monitored distraction parameters for the child.
13. View a graphical report displaying the detected distraction parameters for the monitored child.

System Usability Scale (SUS) Questionnaire Results.

The results analysed in this section correspond to the responses to the SUS questionnaire (see Appendix B.1) provided by the 12 guardians. Based on the collected data, Figure 13 shows a mean SUS score of 81.46%, with a standard deviation of 11.65. According to the adjective ratings (Worst imaginable, Poor, Acceptable, Good, Excellent, Best imaginable) proposed by Bangor, Kortum, and Miller (2008) to qualitatively interpret system usability scores, it is evident that Torddis achieved a “Good” level of usability based on the guardians’ evaluations.

The results of the SUS questionnaire (Brooke, 1996) reflect a positive perception of the system, with scores emphasizing its ease of use and effectiveness. The consistency in users’ responses demonstrates that the interface design and the initial setup processes are properly optimized for users with varying levels of technological experience. This reinforces the robustness of the system as an accessible tool for guardians with different educational backgrounds (Deng et al., 2024).

Results of the Open-Ended Questions Questionnaire.

At the end of the SUS questionnaire, the guardians answered eight open-ended questions (see Appendix B.2) to provide their personal opinions.

The first question was “What is your general opinion about the system?”, to which some guardians responded that it supports students’ concentration, while others highlighted its appealing design, as shown in Figure 14a. The second open-ended question asked was “Did you like the sounds and/or lights used by Torddis?”. Most guardians expressed a favorable opinion, as they believed that light alerts help keep the student awake and were satisfied with the auditory alarms (see Figure 14b).

Figures 14a and 14b summarize these opinions.

Questions 3 (“Do you like the design of the Torddis mobile application? Why?”), 4 (“Is the way the data about your child’s distraction monitoring is displayed in the Torddis mobile app appropriate? Why?”), 5 (“Do you believe this system would help improve your child’s concentration and keep you informed while they do their homework? Why?”), and 7 (“Would you be willing to continue using the Torddis system?”) received entirely positive feedback. The guardians stated that they liked the screen design (graphical interfaces) of the mobile application, praised the organization of historical data and graphical reports, confirmed that the system prototype effectively supports children’s concentration by keeping them informed when distractions occur, and expressed their willingness to continue using the Torddis system.

Question 6 aimed to gather information about possible improvements to Torddis. The feedback included the suggestion to use a higher-quality camera and enhance the camera mounting support. However, a high percentage of users indicated that no further improvements were necessary. These results are illustrated in Figure 14c. Question 8 collected the reasons why guardians would recommend using Torddis for monitoring students during their school-related tasks at home. Figure 14d presents the reasons mentioned by the guardians, highlighting its usefulness in monitoring and its appealing design.

6.8.2 Recognition Analysis

Table 16 presents the average recognition times recorded by Torddis during each test. These average times correspond to the latency in recognizing individuals, facial expressions, drowsiness states, and distracting objects. The number of tests per recognition type ranged from 14 to 32, except for the “Unsuccessful” cases, which occurred in only 1 or 2 trials (depending on the recognition type) where Torddis failed to achieve recognition.

In each trial, the time taken by the system to correctly identify each element was measured in seconds, along with a record of whether the recognition was successful or not.

Although Table 16 includes unsuccessful recognition cases, these were not considered in the main analysis, as the aim was to evaluate response times. Since unsuccessful responses are recorded with a time of 0.00 seconds, their inclusion could bias the results and lead to non-representative statistical values.

Table 16: Aggregated recognition results.

No.	Person Recognition		Facial Expression		Drowsiness Detection		Distracting Objects	
	Latency	Successful?	Latency	Successful?	Latency	Successful?	Latency	Successful?
1	0.60	Yes	1.20	Yes	3.50	Yes	1.68	Yes
2	0.57	Yes	0.82	Yes	2.73	Yes	1.68	Yes
3	0.00	No	0.70	Yes	0.00	No	0.00	No
4	0.78	Yes	1.30	Yes	3.68	Yes	1.79	Yes
5	1.02	Yes	0.78	Yes	2.24	Yes	0.00	No
6	1.20	Yes	0.97	Yes	2.03	Yes	2.05	Yes
7	0.82	Yes	0.87	Yes	3.03	Yes	1.95	Yes
8	0.76	Yes	1.05	Yes	4.09	Yes	1.77	Yes
9	1.20	Yes	0.70	Yes	3.36	Yes	1.91	Yes
10	0.88	Yes	0.00	No	2.63	Yes	1.98	Yes
11	0.53	Yes	1.23	Yes	0.00	No	1.73	Yes
12	0.69	Yes	1.99	Yes	4.69	Yes	4.27	Yes
13	0.67	Yes	1.07	Yes	2.97	Yes	1.77	Yes
14	0.82	Yes	1.83	Yes	3.29	Yes	2.69	Yes
15	0.93	Yes	1.68	Yes	3.14	Yes	2.81	Yes
16	0.58	Yes	1.76	Yes	3.45	Yes	2.94	Yes

Descriptive Analysis of Recognition Latency Times:

Table 17 presents the descriptive results of the latency values recorded by Torddis during its operation. This system, designed to be non-intrusive, collects data in real time, which directly influences the quality and distribution of the recorded measurements. The main findings are discussed below:

- *Person Recognition:* The low data dispersion, evidenced by the small standard deviation and the closeness among the measures of central tendency (mean, median, and mode), indicates that person detection is consistently achieved across users. This consistency is further supported by the narrow range between the minimum and maximum values, as well as a distribution centered around the quartiles. These findings suggest that Torddis reliably captures clear and stable signals related to human presence without interfering with the environment or participants. The observed uniformity reinforces the robustness of this functionality in non-intrusive contexts.
- *Facial Expression Recognition:* This category exhibits greater variability, as reflected in the standard deviation and in a mode that differs significantly from the mean and median. Although the quartiles indicate some degree of concentration, the range

Table 17: Measures of central tendency and dispersion for recognition latency.

	People	Facial Expressions	Sleep Detection	Distracting Objects
Number of observations	15	15	14	14
Mean	0.803	1.197	3.202	2.216
Median	0.780	1.070	3.215	1.930
Mode	0.82 and 1.2	0.700	Amodal	1.68 and 1.77
Standard deviation	0.213	0.431	0.699	0.732
Minimum	0.530	0.700	2.030	1.680
First quartile (Q1)	0.635	0.845	2.790	1.770
Second quartile (Q2)	0.780	1.070	3.215	1.930
Third quartile (Q3)	0.905	1.490	3.488	2.530
Maximum	1.200	1.990	4.690	4.270

between extreme values is broader than in the previous category. This variability may be attributed to individual differences in facial expressions, as well as to environmental factors such as lighting conditions and capture angle. Although latency values remain within acceptable limits, these conditions can affect recognition accuracy in certain cases, suggesting the need to improve facial recognition algorithms without compromising user comfort.

- *Drowsiness Detection:* Latency values in this category are noticeably higher, with elevated mean and median, as well as a broad range and a distribution without a defined mode (amodal). This situation reflects the inherent complexity of detecting subtle signs of drowsiness through non-intrusive methods. The quartiles indicate considerable dispersion, and the standard deviation reinforces this interpretation. While these results are expected, given that drowsiness signals are less evident, the system could benefit in the future from more specialized techniques or sensors to optimize both the speed and accuracy of detection.
- *Detection and Recognition of Distracting Objects:* The wide range of latency values, combined with a high standard deviation and a bimodal mode, indicates considerable variability in the detection of distracting objects. This can be attributed to the diversity of the physical environment, object movement, or lighting conditions—typical variables in uncontrolled settings. Nevertheless, the distribution of the quartiles and the relative proximity between the mean and median suggest that, although complete uniformity is not achieved, the system is capable of recognizing distracting objects in most cases, demonstrating reasonable performance in dynamic and real-world scenarios.

Correlation Between Latency Times:

To compute the correlation matrix, missing values (to complete a total of 16 valid entries) were replaced with the mean latency corresponding to each category. The correlation matrix of latency times for the recognition of people, distracting objects, facial expressions, and drowsiness detection is presented in Table 18. This matrix allows for the examination of linear relationships between the latencies of the different

system modules: *person detection*, *facial expression recognition*, *drowsiness detection*, and *distracting object identification*. It was constructed using Pearson’s correlation coefficient (r), which quantifies the strength and direction of the linear association between pairs of numerical variables.

Table 18: Correlation matrix of recognition module latencies.

	People	Facial Expressions	Drowsiness	Objects
People	1.000	-0.332	-0.426	-0.031
Facial Expressions	-0.332	1.000	0.521	0.744
Drowsiness	-0.426	0.521	1.000	0.476
Objects	-0.031	0.744	0.476	1.000

The analysis reveals a strong positive correlation between the latencies of the *facial expressions* and *distracting objects* modules ($r = 0.744$), suggesting a high degree of interdependence between these processes. This behavior may be attributed to the combined use of deep visual analysis techniques, which involve intensive processing of facial images, or to a shared architecture in the computational implementation that simultaneously evaluates multiple features within the same regions of interest.

A moderate positive correlation is also observed between *facial expressions* and *drowsiness detection* ($r = 0.521$), and between *drowsiness* and *distracting objects* ($r = 0.476$). These coefficients indicate that an increase in latency in one of these modules tends to be associated with proportional increases in the others, which could be linked to the shared computational resource load, such as the use of the graphics coprocessor or common inference libraries.

In contrast, latency in the *person detection* module exhibits moderate negative correlations with both *facial expression recognition* ($r = -0.332$) and *drowsiness detection* ($r = -0.426$). This pattern suggests a possible competition for resources among these modules or a prioritization in the execution flow that reduces latency in one process at the expense of increasing it in another.

Finally, the near-zero correlation between latency in the *person recognition* and *distracting objects* modules ($r = -0.031$) indicates that these modules operate statistically independently in terms of execution time. Such functional independence may be advantageous for optimizing parallel or segmented execution in future iterations of the system.

In summary, this quantitative analysis enables the identification of significant relationships between system components, revealing both synergies and potential performance conflicts, which are essential for enhancing architectural design and operational efficiency in mobile environments.

6.9 Maintenance

This phase has not yet been executed due to the limited timeframe required for the development of each deliverable, as well as the time allocated for the overall evaluation of the project. Nevertheless, the system must be maintained in accordance with the preventive guidelines and the expected lifespan of each of its components, in order to ensure proper functionality. Table 19 outlines some of the potential maintenance tasks. As indicated, each maintenance task—depending on its nature—should be carried out either on a quarterly or semiannual basis, although the actual frequency may vary depending on the operational environment in which the system is deployed.

Table 19: Definition of selected maintenance tasks.

No.	Maintenance Tasks	Quarterly	Semiannually
1	Internal cleaning of the device.	*	
2	Check whether the device components exhibit any imperfections or damage.		*
3	Inspect cable connections, condition of cables, connectors, etc.		*
4	Test the operation of the buzzer and LEDs.	*	
5	Test the functionality of the computing system (web application server, web services, mobile application, IoT device).		*

Preventive maintenance schedules are defined based on the specific component and its expected service life. Likewise, corrective maintenance for each physical component of the system should be performed before the end of its operational lifespan. The expected service life of each component is presented in Table 20. The lifespan of resistors is expected to exceed that of the component with the highest durability in the system.

The estimation of the components' service life has been primarily based on references from scientific literature. However, in the absence of relevant academic sources, the specifications provided by manufacturers and distributors have been considered, supplemented in certain cases with information obtained from specialized forums.

7 Statistical Analysis of the Study

Statistical analysis plays a fundamental role in the understanding and evaluation of systems such as Torddis, enabling the transformation of collected data into robust evidence that supports the study's conclusions. This approach is essential for identifying patterns, relationships, and significant differences among the evaluated variables, ensuring that the results accurately reflect the dynamics observed during the use of the device.

Through descriptive and analytical statistical methods, a rigorous validation of the findings is ensured, highlighting key aspects of the system such as its acceptance by users, its effectiveness in enhancing self-regulation, and its pedagogical

Table 20: Service life of the hardware components used in Torddis.

Component	Service Life (years)	Conditions That Reduce Service Life
ESP32-CAM	20 (ESPRESSIF, 2022)	Exposure to extreme temperatures, high humidity, dust, or mechanical vibrations.
ESP8266 Node MCU	20.83 (Amri & Setiawan, 2018) to 57 (Ardumotica, 2023 ; ESPRESSIF, 2018)	Extreme temperatures, high humidity levels, dust, voltage spikes, power supply noise, frequent power cycling, intensive WiFi usage at maximum power, firmware updates.
Passive Buzzer	3 (Huawha Electronics, 2023)	Excessive voltage, overcurrent, extreme temperatures, vibrations or mechanical shocks, moisture or corrosion, prolonged continuous operation.
LEDs	3 (Casamayor et al., 2015) to 14 (Green Lighting, 2024 ; Shenzhen Cary Technology Co., Ltd, 2024)	Component degradation, manufacturing conditions.
Resistors and others	∞ (expected)	High voltage stress (Simon et al., 2017).

impact. The application of appropriate statistical tools not only reinforces the reliability of the results but also provides a solid foundation for future improvements and implementations of the system in diverse educational contexts.

7.1 Descriptive Statistics

This section presents and analyzes the descriptive statistics obtained from the collected data. These statistics play a crucial role in understanding the behavior of the evaluated variables and their impact on the acceptance and effectiveness of the system.

Table 21 shows measures of central tendency, such as the mean and median, which allow for the identification of general patterns among tutors and students regarding the Torddis system. For instance, the average daily time devoted by tutors to supervising school tasks is 2 hours, indicating active but limited participation due to their time constraints caused by everyday responsibilities. The responses to the statements related to the perception and use of the device (Statements (1) to (7), see Appendix B.3) reveal consistently high average values (ranging from 3.58 to 4.42 on a 5-point Likert scale), suggesting a general acceptance of the system.

On the other hand, dispersion indicators, such as the standard deviation, provide insight into the variability of the responses. The standard deviation for the statements (ranging from 0.67 to 0.98) indicates that participants' perceptions were relatively homogeneous, which reinforces the consistency in the user experience of the device. However, the greater variability in *Statement (3)* suggests that some students may have perceived the device as more intrusive, which could necessitate design adjustments.

Table 21: Descriptive statistics on supervision time and the usage and effectiveness of the developed device.

Statistic	Supervision Time	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mean	2.00	4.42	4.42	2.67	3.58	4.00	4.42	4.42
Median	2.00	5.00	4.00	3.00	4.00	4.00	5.00	5.00
Standard Deviation	0.69	0.90	0.67	0.98	0.90	0.85	0.79	0.79
Minimum	1.00	3.00	3.00	1.00	2.00	3.00	3.00	3.00
Maximum	3.00	5.00	5.00	4.00	5.00	5.00	5.00	5.00

Items (n) with $n = 1 \dots 7$ correspond to the seven statements evaluated on a 5-point Likert scale related to the prototype's effectiveness (see Appendix B.3, Table B3). *Supervision time* represents the number of daily hours that tutors dedicate to supervising their children's schoolwork.

Likewise, the minimum and maximum values reveal possible differences in participants' experiences, highlighting areas that may require improvement. For instance, while some tutors reported dedicating only one hour to supervision, others reached up to three hours, which could influence the responses related to the system's effectiveness.

These results make it possible to identify not only the general level of acceptance of the system but also specific areas that could benefit from targeted future interventions. Taken together, the descriptive statistics provide a foundation for validating the device's design and supporting the discussion regarding its pedagogical and practical potential.

7.2 Normality Analysis of the Data

To determine whether the numerical variables follow a normal distribution, two complementary tests were applied: the Shapiro–Wilk (SW) test and the Kolmogorov–Smirnov (KS) test (Correa-Álvarez, Rojas-Mora, Zumaqué Ballesteros, & Bru-Cordero, 2025). The SW test is highly sensitive to deviations from normality and is particularly suitable for small sample sizes, whereas the KS test compares the empirical distribution of the data with a theoretical distribution (in this case, the normal distribution) and is more robust in the presence of heterogeneous variance or larger sample sizes (Kobayashi, Pillai, Ramana, & Thomas, 2025). Both tests were employed to ensure a comprehensive analysis by evaluating different aspects of normality in the data.

The results presented in Table 22 show that the SW and KS tests do not always produce consistent conclusions, as they assess different properties of the normality assumption (Correa-Álvarez et al., 2025). Therefore, the results of both tests were interpreted in a complementary manner to achieve a more robust evaluation of the normality of the dataset.

The *p-value* shown in Table 22 for both tests indicates whether the null hypothesis is accepted or rejected; that is, whether the variables follow a normal distribution or not. The columns *SW Conclusion* and *KS Conclusion* display the results of the

Table 22: Results of the normality analysis for tutors' responses to the effectiveness questions

Variable	SW p -value	SW Conclusion	KS p -value	KS Conclusion
Tutor's Age	0.2277	✓	0.9289	✓
Supervision Time	0.0119	×	0.1885	✓
Student's Age	0.1736	✓	0.6257	✓
Statement (1)	0.0016	×	0.0775	✓
Statement (2)	0.0063	×	0.2562	✓
Statement (3)	0.0528	✓	0.1898	✓
Statement (4)	0.0598	✓	0.2081	✓
Statement (5)	0.0199	×	0.3779	✓
Statement (6)	0.0016	×	0.0775	✓
Statement (7)	0.0016	×	0.0775	✓

×: Does not follow a normal distribution. ✓: Follows a normal distribution.
SW = Shapiro-Wilk. KS = Kolmogorov-Smirnov.

hypothesis testing (H_0 : The variable follows a normal distribution. H_1 : The variable does not follow a normal distribution).

While the KS test indicates that all variables follow a normal distribution, the SW test suggests that some variables conform to normality, whereas others do not. To ensure the validity of subsequent analyses, appropriate statistical tests were selected based on these results. Nevertheless, the Pearson correlation coefficient test was applied to all variables.

7.3 Correlation between Statement (5) and the Other Variables

Pearson correlation tests were employed to explore significant relationships and differences among the variables under study and their corresponding factors. Considering all variables as normally distributed, the Pearson correlation analysis enabled the identification of relationships between the variable “*Statement (5): After discontinuing use of the device, the student demonstrated improvement in self-regulation and concentration while completing homework at home*” and the remaining numerical variables. The correlations observed were statistically significant in several cases, highlighting important patterns regarding the impact of the device, as shown in Table 23.

Firstly, the variables associated with the following statements—**Statement (1)**: *The device was perceived by the student as appealing*, **Statement (3)**: *The student did not perceive the presence of the device as intrusive*, **Statement (4)**: *The device was used throughout the effective time devoted to performing school tasks*, **Statement (6)**: *Although the device usage time was short, it is expected to significantly contribute to task completion and the academic performance of the student*, and **Statement (7)**:

Table 23: Correlations between Statement (5) and the Other Variables

Variable	Correlation with Statement (5)	p-value	Type of Relationship
Age of the tutor ^{sw}	-0.2725	0.3916	△
Supervision time	-0.1841	0.5668	△
Age of the student ^{sw}	-0.1407	0.6627	△
Statement (1)	0.7762	0.0030	★
Statement (2)	0.5669	0.0546	✓
Statement (3) ^{sw}	0.7500	0.0050	★
Statement (4) ^{sw}	0.8165	0.0012	★
Statement (6)	0.7762	0.0030	★
Statement (7)	0.7762	0.0030	★

^{sw} Normal variables according to the Shapiro-Wilk test.

△: Weak. ✓: Strong. ★: Very Strong.

You would be willing to continue using the device—showed *very strong* positive correlations with *Statement (5)*, with values ranging from 0.7500 to 0.8165. In addition, **Statement (2): The mobile application accurately recorded the events that occurred**, yielded a strong correlation (p-value = 0.5669).

Although age is often regarded as an indicator of greater responsibility (Moss & Wilson, 2018), in this study, the variables *Age of the tutor* and *Age of the student* exhibited weak and non-significant correlations, with values of **-0.2725** and **-0.1407**, respectively. These findings suggest that the perceived effectiveness of the device is not influenced by the age of either tutors or students. This result highlights the versatility of the system, demonstrating its applicability across diverse demographic groups without being constrained by age-related factors.

Finally, the results underscore that characteristics directly related to device usage—such as its design and operational effectiveness—have a greater impact on improving self-regulation than demographic factors. This provides strong evidence that user-centered design is a key strategy in the development of technology-based educational systems.

7.4 ANOVA Analysis

The ANOVA analysis conducted on the variables related to *Statement (5)* revealed significant differences in several statements associated with the effectiveness of the Torddis device, while other variables did not exhibit notable differences. The results presented in Table 24 have important implications for assessing the system’s impact across different dimensions.

Table 24: ANOVA analysis results

Variable	F-Statistic	p-value	Conclusion
Tutor’s Age ^{sw}	0.8162	0.4723	×
Supervision Time	0.2411	0.7907	×
Student’s Age ^{sw}	1.6419	0.2467	×
Statement (1)	11.0625	0.0038	★
Statement (2)	2.9118	0.1059	NS
Statement (3) ^{sw}	5.7857	0.0242	✓
Statement (4) ^{sw}	10.6875	0.0042	★
Statement (6)	11.0625	0.0038	★
Statement (7)	11.0625	0.0038	★

^{sw} Variables following a normal distribution according to the Shapiro–Wilk test.
 ×: No significant differences. NS: Not significant. ✓: Significant difference. ★: Highly significant difference.

Statement (1), which assesses the perceived attractiveness of the device, along with Statements (4), (6), and (7), which address aspects related to the use of the device, exhibited *highly significant differences*, thereby reinforcing the system’s versatility and effectiveness. Statement (3), related to the discretion of the device, presented *p*-values below 0.05, indicating statistically significant differences among the evaluated groups. This supports the notion that the device’s design has a differential impact on its acceptance and perception, potentially influenced by contextual factors such as the age or environment of the learner.

On the other hand, variables such as the tutor’s age, the student’s age, and supervision time did not show statistically significant differences ($p > 0.05$), suggesting that the impact of the Torddis system is not conditioned by these demographic variables. This further reinforces the notion that the system is inclusive and adaptable to a wide range of user profiles.

7.5 Post Hoc Tests for Variables with Significant Differences

Post Hoc tests (Ogrodniczek, Rudawska, & Müller, 2025) were conducted to identify the specific groups in which statistically significant differences were observed in the ANOVA results presented in Table 24:

- **Statement (1):** The results shown in Table 25 revealed statistically significant differences between groups 3 and 4 ($p = 0.0079$), and between groups 3 and 5

($p=0.0049$). This indicates that the attractiveness of the device is perceived differently across some user subgroups, which may be related to individual or contextual preferences.

Table 25: Post hoc tests for 'Statement (1)'.

Group 1	Group 2	Mean Difference	p-value	Lower Bound	Upper Bound	Rejected
3	4	1.3333	0.0079	0.4027	2.264	Yes
3	5	1.6667	0.0049	0.5920	2.7413	Yes
4	5	0.3333	0.5951	-0.5973	1.264	No

- **Statement (3):** Although this statement exhibited statistically significant differences in the ANOVA analysis, the Post Hoc tests identified significance only between groups 3 and 5 ($p = 0.0194$), as shown in Table 26. This suggests that not all subgroups perceive the discretion of the device in the same manner.

Table 26: Post hoc tests for 'Statement (3)'.

Group 1	Group 2	Mean Difference	p-value	Lower Bound	Upper Bound	Reject H_0
3	4	1.0000	0.1769	-0.4216	2.4216	No
3	5	2.0000	0.0194	0.3585	3.6415	Yes
4	5	1.0000	0.1769	-0.4216	2.4216	No

- **Statement (4):** Statistically significant differences were observed between Groups 3 and 4 ($p=0.0176$), and between Groups 3 and 5 ($p=0.0038$), as shown in Table 27. This indicates that the effective use of the device varies depending on the group, which may be related to factors such as the level of supervision or environmental conditions.

Table 27: Post hoc tests for 'Statement (4)'.

Group 1	Group 2	Mean Difference	p-value	Lower Bound	Upper Bound	Rejected
3	4	1.3333	0.0176	0.2587	2.408	Yes
3	5	2.0000	0.0038	0.7591	3.2409	Yes
4	5	0.6667	0.2461	-0.408	1.7413	No

- **Statements (6) and (7):** The Post Hoc tests revealed patterns similar to those observed in Statement (1), showing significant differences between Groups 3 and 4, as well as between Groups 3 and 5. These findings support the positive perception regarding the device's impact on self-regulation and the intention to continue its use.

These results highlight that Torddis is an effective and versatile tool, whose impact may vary depending on contextual factors such as the environment and user preferences. The identification of significant differences among specific groups underscores the need to tailor certain aspects of the device's design and implementation in order to maximize its effectiveness across diverse educational settings.

Table 28: Post hoc tests for 'Statement (6) and (7)'

Group 1	Group 2	Mean Difference	p-value	Lower Bound	Upper Bound	Reject
3	4	1.3333	0.0079	0.4027	2.264	Yes
3	5	1.6667	0.0049	0.5920	2.7413	Yes
4	5	0.3333	0.5951	-0.5973	1.264	No

Overall, the statements related to the use and acceptance of the device proved to be relevant in the evaluation of Torddis, while demographic and contextual variables did not significantly condition its effectiveness. This reinforces the system's robustness and its general applicability in a wide range of user profiles and learning environments.

7.6 Tests for Variables That Do Not Follow a Normal Distribution According to the SW Test

For the variables *Supervision time*, *Statement (1)*, *Statement (2)*, *Statement (6)*, and *Statement (7)*, whose p-value was less than 0.05 in the Shapiro-Wilk test, non-parametric tests were applied (Smeeton, Spencer, & Sprent, 2025). In this case, Spearman's rank correlation coefficient (Yu & Hutson, 2024b) was used to analyze relationships between these variables, and the Kruskal-Wallis (KW) test (Jamil & Khanam, 2024) was applied to both categorical variables and numerical variables that did not follow a normal distribution, in order to assess significant differences among groups.

Table 29 presents the results from applying Spearman's correlation coefficient and the KW test, showing the observed differences among the variables in relation to *Statement (5)*, which assesses the perceived improvement in self-regulation and concentration of the tutee after using the Torddis system. A detailed analysis of the meaning of these results is provided below.

Categorical variables such as *Tutor's Education Level*, *Tutor's Gender*, *Area of Residence*, *Occupation*, and *Type of Internet Signal* did not exhibit significant differences according to the Kruskal-Wallis test ($\emptyset$), indicating that these characteristics do not significantly influence the perceived effectiveness of the device. Furthermore, their Spearman correlation coefficients ranged from -0.2500 to 0.0000, reinforcing the absence of significant relationships with the response variable.

In contrast, *Statement (1)*, *Statement (6)*, and *Statement (7)* showed significant differences ($\checkmark$) and strong correlations ($\star$) with *Statement (5)*. These results suggest that positive perceptions regarding the device's usefulness, willingness to continue using it, and its impact on completing academic tasks are directly related to the perceived improvement in the tutee's self-regulation and concentration. This finding underscores the relevance of these variables in evaluating the effectiveness of **Torddis**.

On the other hand, *Supervision Time* and *Statement (2)* showed weak correlations (Δ) and no significant differences ($\emptyset$ and $\triangleleft$), suggesting a limited effect of these variables on the perceived improvement in self-regulation. These results indicate that although such variables may exert some influence, they are not decisive in shaping the perception of the system's effectiveness.

Table 29: Results of Kruskal-Wallis and Spearman correlation Tests with respect to Statement (5)

Variables	Spearman $Correlation$	Spearman $p-value$	KW $Statistic$	KW $p-value$	Conclusions
Tutor's Education Level	0.0000	1.0000	0.0000	1.0000	$\times \emptyset$
Tutor's Gender	-0.2500	0.4332	0.6875	0.4070	$\Delta \emptyset$
Area of Residence	0.0000	1.0000	0.0000	1.0000	$\times \emptyset$
Occupation Type	0.0000	1.0000	0.0000	1.0000	$\times \emptyset$
Type of Internet Signal	-0.2500	0.4332	0.6875	0.4070	$\Delta \emptyset$
Supervision Time	0.1925	0.5490	0.8730	0.6463	$\Delta \triangleleft$
Statement (1)	0.7698	0.0034	6.6349	0.0362	$\star \checkmark$
Statement (2)	0.5508	0.0635	3.6056	0.1648	$\Delta \times$
Statement (6)	0.7698	0.0034	6.6349	0.0362	$\star \checkmark$
Statement (7)	0.7698	0.0034	6.6349	0.0362	$\star \checkmark$

$\times$: No correlation. Δ : Weak correlation. $\triangle$: Moderate correlation. $\star$: Strong correlation. $\emptyset$: No significant difference. $\triangleleft$: No significant effect. $\checkmark$: Significant difference.

7.7 Consolidation of the Technical and Pedagogical Discussion

The Torddis system represents an innovative fusion of technical advances and pedagogical principles, demonstrating its effectiveness in enhancing the user experience in supervising and supporting students' academic development at home. This integration is manifested in several key aspects evident both in the system's outcomes and in its design.

From a technical standpoint, Torddis employs advanced computer vision algorithms to detect distractions in real time, such as facial expressions indicative of inattention, the use of distracting objects, and signs of drowsiness. These capabilities are embedded in an IoT device that operates non-intrusively, ensuring that the system's interventions do not disrupt the natural flow of learning at home (Asish et al., 2022; E. Chen, Tseng, & Lo, 2022; Roy, Mary, C, & Selvi, 2022; Uçar & Özdemir, 2022). Such techniques have proven effective in contexts such as attention monitoring during

online classes and distraction management through real-time alerts using computer vision and IoT-based tools (E. Chen et al., 2022; Uçar & Özdemir, 2022). Additionally, the system’s ability to issue customizable visual and auditory alerts enables students to redirect their attention without requiring constant supervision from a tutor (Ackermans et al., 2025). This supports the development of self-regulation skills, such as the ability to identify distractions and adopt strategies to autonomously correct them (X. Wang et al., 2025).

From a pedagogical perspective, the design of Torddis aligns with contemporary theories of autonomous learning and self-regulated education. The visual and auditory alerts serve not only as reminders but also as pedagogical tools that motivate students to stay focused on their academic tasks (Mohamed & Lamia, 2018). For instance, the system provides immediate feedback to students, which encourages active reflection on their performance (Ackermans et al., 2025). This is complemented by the historical analysis of behavioral patterns, which allows tutors to identify specific areas where students may need additional support, such as time management or maintaining focus in noisy environments.

The combined impact of these technical and pedagogical features is reflected in the system outcomes, where tutors have reported significant improvements in student concentration. The system’s usability score of 81.46% on the SUS scale demonstrates positive acceptance of Torddis by end users. Moreover, the inclusion of statistical charts displaying students’ event history in the mobile application not only facilitates effective monitoring but also empowers tutors by providing concrete data to make informed decisions regarding educational supervision (X. Wang et al., 2025).

Lastly, the system’s design as a customizable IoT device, coupled with an intuitive mobile application, ensures a seamless and accessible user experience. This approach not only enhances learning in home environments but also strengthens the student–tutor relationship by promoting a collaborative learning process. Thus, Torddis not only addresses the technical requirements of a real-time supervision system but also establishes a robust pedagogical framework to support students’ holistic development in their autonomous learning journey (Li et al., 2024).

7.8 Educational Impact of the System

The Torddis system has proven to be a highly promising educational tool, particularly in aspects related to attention, self-regulation, and academic performance monitoring within home environments. Evidence collected through qualitative responses from tutors highlights the perceived usefulness of the system in promoting focus on school tasks and minimizing distractions. Through statements such as “Yes, because it alerts when the child is distracted” or “Yes, to know what they are doing and call their attention,” users clearly express how the system facilitates the monitoring of children’s behavior, aligning with contemporary approaches to self-regulated learning (Zimmerman, 2000).

The willingness of tutors to continue using the system represents a second key indicator of its educational impact. When asked, “Would you be willing to continue using the Torddis system?”, the responses were not only affirmative, but also emotionally and functionally expressive: “Yes, very much so,” “Yes, it would help me be

more attentive to my child,” or “Yes, to know what is going on with my child.” This sustained acceptance of the system, beyond its novelty, indicates a process of technological appropriation that reinforces its value as a pedagogical resource and suggests that it fulfills a concrete function in the home learning dynamic. Within the framework of the Technology Acceptance Model (Davis, 1989), such responses may be interpreted as a strong perception of usefulness.

Even more revealing is the fact that tutors are willing to recommend the system to others. Expressions such as “Yes, so their children stay focused on their homework” or “Yes, so parents know what their children are doing” reflect a social validation of the system, indicating that its educational value has been sufficiently recognized by those who have experienced it. This form of recommendation, which goes beyond individual experience, constitutes an indicator of positive impact with the potential for transfer to other educational contexts, validating the system’s social relevance and scalability (Panadero, 2017).

Taken together, these findings offer a coherent and in-depth view of the educational impact of the Torddis system. It is not merely a technological monitoring tool, but rather a resource perceived as valuable for fostering study habits, enhancing concentration, and strengthening communication between parents and children. The explicit support of tutors—manifested both in their willingness to continue using the system and in their inclination to recommend it—reinforces its pedagogical validity. This body of evidence constitutes a robust qualitative foundation that justifies its implementation in households as a technological mechanism that enables parents to supervise their children while they autonomously carry out their school tasks.

8 Conclusions and Future Work

This study evaluated the capabilities of Torddis to monitor children’s distractions while engaging in school activities at home. Torddis combines advanced IoT and AI technologies, such as smart cameras and proximity sensors, to provide proactive real-time monitoring in domestic environments.

In terms of **technological design**, Torddis integrates facial recognition, detection of distracting objects, and drowsiness analysis in an IoT environment, enabling proactive supervision through built-in visual and audio alerts. This approach contrasts with previous works, such as Campbell (2015) and Uçar and Özdemir (2022), which are limited to controlled environments and lack adaptability to household settings. Similarly, studies such as Argel et al. (2023) and Nguyen et al. (2019) employ computer vision for distraction monitoring, but do not include real-time adaptive intervention capabilities—a key feature of Torddis. Moreover, the TDDM4IoTS methodology ensures an iterative and robust development process, aligned with both technical and pedagogical quality standards, as recommended by studies like [reference anonymised], Hachad et al. (2020), and Salter (2014).

Studies such as [reference anonymised] and Nguyen et al. (2019) present systems that identify distractions in classrooms using facial recognition, but their outcomes are limited to data collection without directly intervening in student behavior. Torddis addresses this gap by offering personalized visual and auditory alerts to redirect

children’s attention. Similarly, Torddis goes beyond the contributions of [Ozdamli et al. \(2022\)](#) and [de Villiers Rader et al. \(2021\)](#), who integrate educational technologies to monitor emotions and gestures, but without adapting to domestic contexts.

With regard to the **monitored parameters** and application context, Torddis incorporates mechanisms to detect facial expressions, distracting objects, and drowsiness. This surpasses the limited range of parameters used in other systems, which focus solely on gestures or posture. For instance, [Campbell \(2015\)](#) uses multispectral imaging to identify distractions caused by mobile devices, whereas [Uçar and Özdemir \(2022\)](#) focuses on head posture to assess engagement in virtual classrooms. Both studies are designed for highly controlled settings, such as classrooms or laboratories, and lack the adaptability required for unstructured home environments. Conversely, [Da Costa Rocha et al. \(2023\)](#) uses robotic technologies to support autonomous learning in urban settings, and [Boumiza et al. \(2017\)](#) implements facial recognition to personalize educational experiences, but without the depth of contextual analysis offered by Torddis.

The **user groups** also differ significantly between related works and Torddis. While most studies focus on students in traditional educational environments, Torddis is specifically designed for children doing schoolwork at home under remote supervision by parents or guardians. This demographic—school-aged children in home-based learning—is traditionally underserved yet critical to the development of autonomous learning skills and presents unique challenges due to the lack of structured supervision. Studies such as [Roberts \(2020\)](#) and [Muljana \(2022\)](#) emphasize the importance of self-regulation strategies in school contexts but do not incorporate adaptive or proactive intervention components.

From a **pedagogical perspective**, Torddis fosters self-regulation and autonomous learning by providing immediate feedback and strategies to improve focus. This functionality is key to enhancing time management and attention, as evidenced in previous educational technology research by [Palmer et al. \(2022\)](#), [Roberts \(2020\)](#), and [C.-H. Wang et al. \(2022\)](#). However, unlike unidimensional tools focused exclusively on monitoring, Torddis includes interactive components that engage tutors, reinforcing the link between technology and human supervision. This approach aligns with pedagogical strategies based on personalization, as discussed by [Salter \(2014\)](#), [Bembich et al. \(2016\)](#), and [Adcroft \(2018\)](#).

In terms of **outcomes**, Torddis not only identifies distractions but also provides adaptive strategies involving both the student and the tutor. This approach is unique compared to works such as [James and Nettikadan \(2019\)](#) and [Ozdamli et al. \(2022\)](#), which limit their scope to administrative supervision, such as attendance tracking or exam cheating detection. On the other hand, studies like [Palmer et al. \(2022\)](#), [Muljana \(2022\)](#), and [Hartley et al. \(2022\)](#) highlight the promotion of self-regulation but rely entirely on active student participation, without integrating automated mechanisms such as those in Torddis to redirect attention in real time. Additionally, Torddis addresses the limitations found in [reference anonymised], which focuses on facilitating classroom interaction but lacks an approach centred on individualized feedback.

The results of the present study indicate that tutors perceive the system as highly useful and effective, with an average SUS score of 81.46%. This reinforces the applicability of Torddis as an academic supervision tool and its potential to positively influence children’s performance and concentration. However, areas for improvement were identified, such as camera quality and the extension of the system to diverse cultural and socioeconomic contexts—issues also noted as limitations in similar systems like those of [Bembich et al. \(2016\)](#) and [Muljana \(2022\)](#). In line with recent proposals such as those by [Narkhede et al. \(2023\)](#) and [Uçar and Özdemir \(2022\)](#), Torddis could integrate deeper analyses of the relationship between distractions and students’ physical environments.

Limitations and Future Recommendations: The limited sample size and short study duration constrain the generalizability of the results. It is recommended to expand the sample in terms of size and geographical diversity and to extend the evaluation period to assess long-term impacts on skills such as self-regulation. Likewise, future system iterations should explore more robust hardware components and cost-effective maintenance solutions to ensure technological sustainability. Furthermore, it is suggested to integrate broader objective metrics, such as biometric sensors and behavioural pattern analysis, to enhance the system’s accuracy, in line with the proposals of [de Villiers Rader et al. \(2021\)](#), [Hachad et al. \(2020\)](#), and [Ozdamli et al. \(2022\)](#).

In *conclusion*, Torddis represents a significant contribution to the field of technology-assisted education by offering a comprehensive and adaptive solution for unsupervised learning environments. Its ability to detect and mitigate distractions in real time, combined with its focus on personalization and self-regulation, highlights its relevance and potential impact in the educational and child supervision domains. This work lays the foundation for future research in the development of IoT and AI technologies applied to autonomous learning, promoting a more inclusive and effective educational environment.

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Statements and Declarations

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Ethics Approval and Consent to Participate. The research project underlying this study was reviewed and approved by the Research Committee of the host institution and subsequently endorsed by the corresponding evaluation tribunal that oversaw the project’s execution and dissemination. The host institution does not operate an institutional ethics committee with jurisdiction over non-clinical educational technology studies of this nature; the referenced academic bodies acted as the

competent oversight instances. The study was conducted in accordance with the ethical principles of the 1964 Declaration of Helsinki and its later amendments. Prior to any data collection, written informed consent was obtained from every parent or legal guardian of the participating children, as reported in the Materials and Methods section of this manuscript; children additionally provided verbal assent before each session. Participation was voluntary and could be discontinued at any moment without consequence for the family.

Consent for Publication. The written informed consent obtained from the participating parents and legal guardians included permission for the anonymised use of their questionnaire responses and of the system-generated event data for scientific publication. The figures included in this manuscript that depict evaluation environments were prepared so that participants are not individually identifiable.

Data Availability. The datasets generated and analysed during the current study — comprising the demographic survey, the ten-item System Usability Scale (SUS) responses, the seven-item Likert effectiveness questionnaire, the open-ended responses and the system-generated event logs (alert frequency and active supervision time) — are available from the corresponding author on reasonable request, subject to the confidentiality restrictions established with the participating families.

Materials Availability. All hardware components required to reproduce the Torddis device (ESP32-CAM module, NodeMCU ESP8266 v3 board, passive buzzer and 5 mm LEDs) are commercial off-the-shelf items whose specifications, alternatives and technical remarks are fully documented in the Hardware Resources subsection of this manuscript. The physical prototype is available at the authors’ laboratory for on-site demonstration on reasonable request.

Code Availability. The mobile application, the firmware for the IoT device and the auxiliary scripts developed during the study are available from the corresponding author on reasonable request for non-commercial academic use. The location of the auxiliary web-based tool used during development has been removed for double-blind review and will be restored in the final version if the manuscript is accepted.

Author Contributions. Author contribution details have been removed for double-blind review. All authors read and approved the final version of the manuscript.

Use of Generative AI and AI-Assisted Technologies. During the preparation of this work, the authors did not use any generative artificial intelligence or AI-assisted technology to create or alter the scientific content, results, figures or conclusions. The authors take full responsibility for the content of the publication.

Appendix A Demographic Survey

1. Full Name: _____

2. Age: _____

3. Education Level:

- No formal education ()

- Primary ()
- Secondary ()
- Higher education ()

4. Gender:

- Male ()
- Female ()
- Other ()

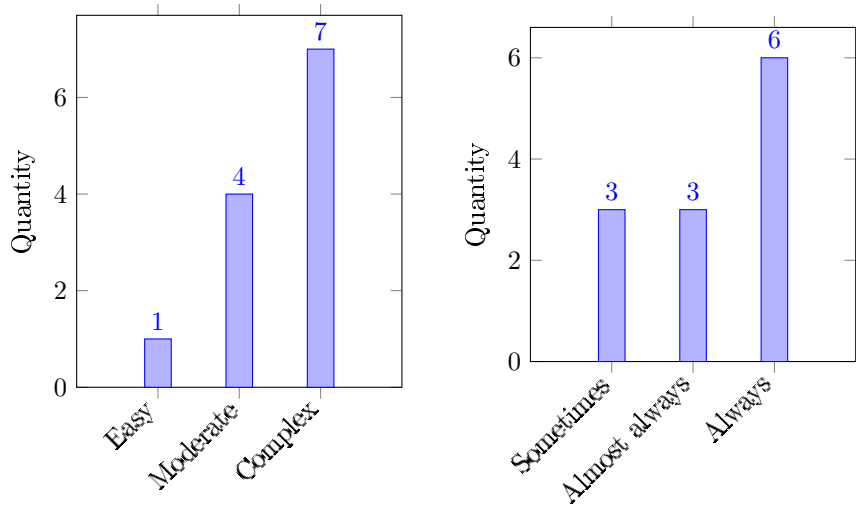
5. How would you describe your experience monitoring your child's distractions while doing schoolwork?

- Easy ()
- Average ()
- Challenging ()
- Very challenging ()

6. How frequently do you monitor your child's academic activities?

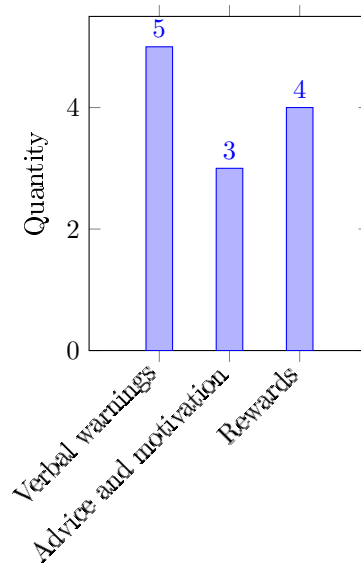
- Never ()
- Sometimes ()
- Almost always ()
- Always ()

7. What strategies have you implemented to improve your child's concentration during schoolwork?



(a) Perceived difficulty of the monitoring process.

(b) Frequency of student monitoring.



(c) Strategies used to improve concentration.

Fig. 12: Responses from guardians to preliminary questions regarding the monitoring of their elementary school children while completing homework at home.

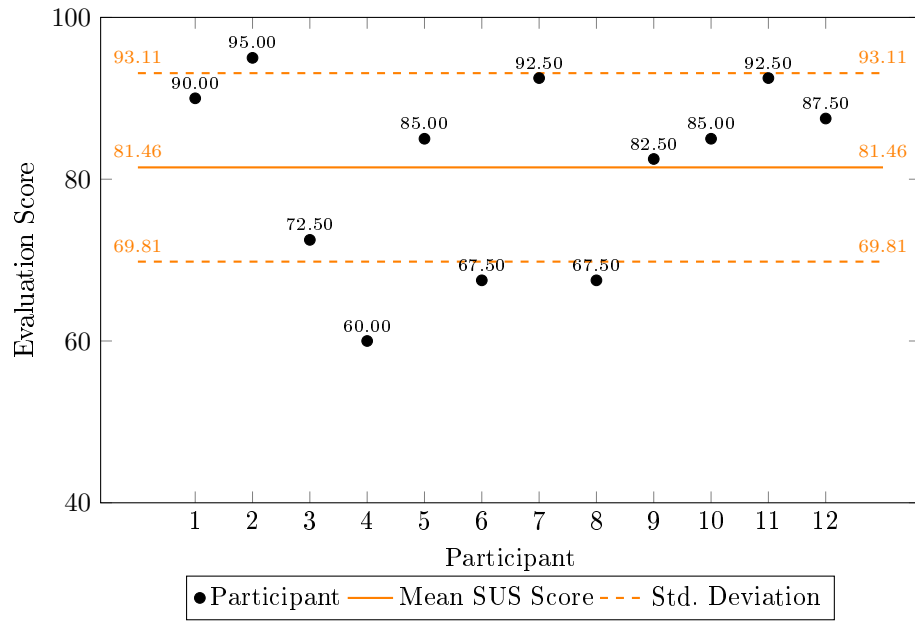
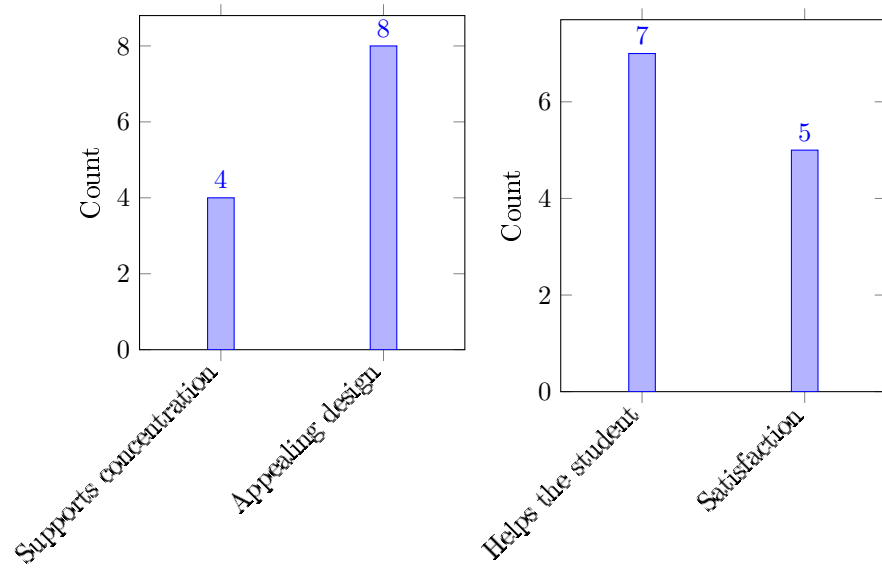
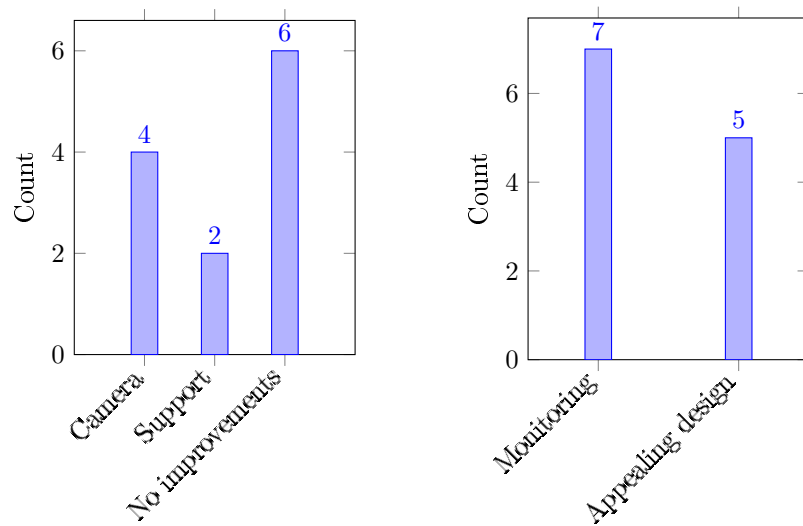


Fig. 13: Mean and standard deviation of guardians' responses to the SUS questionnaire.



(a) General opinion about the Torddis system. (b) Auditory and visual alerts of the system.



(c) Suggestions for improving the system. (d) Reasons to recommend the system.

Fig. 14: Results of guardians' responses to open-ended questions regarding the monitoring process of their primary school children using Torddis while performing academic tasks at home.

Appendix B Questionnaires to Evaluate the Usability and Effectiveness of Torddis

This appendix compiles the instruments used to assess both the usability and effectiveness of the Torddis system during its pilot implementation. The questionnaires were designed following validated models such as the System Usability Scale (SUS) and adapted to the educational and domestic context in which the prototype was deployed. The collected responses contribute significantly to understanding user satisfaction, perceived utility, and the potential pedagogical impact of the system. This section includes three types of questions: a Likert-scale questionnaire to measure usability, a set of open-ended questions to capture qualitative insights, and a final Likert-based instrument focused on evaluating the system’s effectiveness in real-world use.

B.1 Likert Scale Questionnaire

The Likert-scale questionnaire is based on the internationally recognized System Usability Scale (SUS), which provides a standardized framework to evaluate the usability of interactive systems. This instrument was selected due to its reliability and widespread application in human-computer interaction studies. The SUS allows capturing both positive and negative perceptions regarding the ease of use, complexity, integration, and learnability of the system. In the context of Torddis, the questionnaire was adapted to reflect specific aspects of its use by guardians in a home-based educational environment. The following items were presented to participants after a defined period of system interaction. Table B1 lists the questions, and respondents were asked to indicate their agreement level using the following scale:

1. Strongly disagree
2. Disagree
3. Neutral (neither agree nor disagree)
4. Agree
5. Strongly agree

B.2 Open-Ended Questions from the SUS Questionnaire

In addition to the structured Likert-scale items, a set of open-ended questions was included to capture more nuanced and personalized perspectives from the users. These questions allowed participants to express their opinions, preferences, and suggestions freely, thereby enriching the quantitative data with qualitative insights. The open-ended format is particularly valuable in educational technology research, as it reveals motivations, emotional responses, and contextual factors that may not be evident through closed questions. Table B2 presents the open-ended questions aimed at deepening the understanding of user experience with both the Torddis device and mobile application.

Table B1: SUS questionnaire

No.	Questions	1	2	3	4	5
1	I would like to use the Torddis system frequently.					
2	I found the system unnecessarily complex.					
3	I thought the system was easy to use.					
4	I would need technical or instructor support to use the system.					
5	I found the various system functions to be well integrated (forming a coherent whole).					
6	I thought the system had too many inconsistencies.					
7	I imagine most people would learn to use the system very quickly.					
8	I found the system very difficult to use.					
9	I felt very confident/comfortable using the system.					
10	I needed to learn many things before I could use the system.					

1: Strongly disagree. 2: Disagree. 3: Neutral (neither disagree nor agree). 4: Agree. 5: Strongly agree.

Table B2: Open-ended questions from the SUS questionnaire

No.	Question
1	In general, what is your opinion about the system?
2	Did you like the sounds and/or lights included in the Torddis device? Please answer yes or no and provide a reason.
3	Did you like the screen design of the Torddis mobile application? Please answer yes or no and provide a reason.
4	Is the way data is displayed about your child's distraction monitoring in the Torddis mobile app appropriate? Why?
5	Do you believe this system could help improve your child's concentration and keep you informed when they are distracted? Please answer yes or no and provide a reason.
6	Do you think there is anything that should be improved in the Torddis system (Device and Mobile App)? If so, what?
7	Would you be willing to continue using the Torddis system? Please answer yes or no and provide a reason.
8	Would you recommend this system to others interested in monitoring children's distraction while doing schoolwork? Please answer yes or no and provide a reason.

B.3 Questions to Assess the Effectiveness of the Prototype

The following section presents a dedicated Likert-scale questionnaire focused on assessing the perceived effectiveness of the Torddis prototype in real educational contexts. These items were specifically designed to evaluate the functionality, relevance, and pedagogical contribution of the system after its application in home environments. Unlike the general usability assessment, this questionnaire targets outcomes such as attention improvement, self-regulation, system attractiveness, and user engagement. Responses to these questions provide empirical evidence regarding the system's practical value, contributing to its validation as a meaningful educational technology. Table B3 presents the list of effectiveness-related items.

Table B3: Likert scale questions to assess system effectiveness

No.	Questions	1	2	3	4	5
1	The device was perceived as attractive by the child.					
2	The mobile application correctly recorded the events that occurred.					
3	The child did not perceive the presence of the device as intrusive.					
4	The device was used throughout the entire time dedicated to school-work.					
5	After discontinuing the use of the device, the child demonstrated improvement in self-regulation and concentration while completing tasks.					
6	Even though the device was used for a short period, it is expected to significantly contribute to task completion and academic achievement.					
7	You would be willing to continue using the device.					

1: Strongly disagree. 2: Disagree. 3: Neutral (neither disagree nor agree). 4: Agree. 5: Strongly agree.

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